MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

**Project:** **Mansourah Massarah Gold Project**

**Customer:** **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

**Title:** **CALCULATION**

**49PR POWER PLANT GUARD ROOM**

**DESIGN CALCULATION OF GUARD ROOM**

**AS BUILT**

**MD-216-49PR-EG-CV-CAL-2014**

| Rev | Date | Revision Text | Prepared By | Reviewed By | Approved By | Employer |
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| D01 | 26-10-2023 | AS BUILT | SHD | SGC | PRI | |
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| A01 | 03-07-2020 | FIRST ISSUE | SHD | SGC | PRI | |

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|---|---|
| Global Project ID | 900707 |
| Delivery Type | ENGINEERING PROCUREMENT CONSTRUCTION |
| Project Name | MANSOURAH MASSARAH GOLD PROJECT |
| Customer | MA'ADEN GOLD AND BASE METALS (MA'ADEN) |
| Project Country | KINGDOM OF SAUDI ARABIA |

| Ma'aden Revision No: | A01 | C01 | D01 |
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| Checked | SGC | SGC | SGC |
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**REFERENCE DOCUMENTS**

| DOCUMENT NUMBER | DOCUMENT DESCRIPTION |
|---|---|
| MD-216-1000-EG-ST-CRT-1001 | Civil Structural Design Criteria |
| MD-216-3000-EG-CV-RPT-1008 | Geotechnical Report |
| SBC 301 :2007 | Saudi Structural - Loading & Forces Code |
| SBC 304 :2018 | Saudi Concrete Structures Code |

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| Approved | PRI | PRI | PRI | | | | Approved | PRI | PRI | PRI | | | |

## DEFINITIONS AND ABBREVIATIONS

| | |
|---|---|
| Employer | MA'ADEN GOLD AND BASE METALS or their representatives |
| EPC Contractor | Consortium of Outotec (OT) and Larsen & Toubro (L&T) |
| Project management Company (PMC) | BECHTEL appointed by EMPLOYER |
| Supplier | An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor |
| Sub- supplier | A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor |
| Shall | Is to be understood as mandatory in relation to the requirements of this Document |
| Should | Is to be understood as a strong recommendation to comply with the requirements of this Document |
| LRFD | Load and Resistance Factor Design |
| F.O.S | Factor of Safety |

| | |
|---|---|
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| Project Country | **KINGDOM OF SAUDI ARABIA** |

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| Approved | PRI | PRI | PRI | | | | Approved | PRI | PRI | PRI | | |

## 1. INTENT

The document is prepared for Design Calculation of Power Plant Guard Room

## 2. SCOPE

The document is prepared for Analysis & Structural Design Verification for Power Plant Guard Room
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
All the Reinforced Concrete members are designed in STAAD Pro using Factored Load Combinations from SBC 304:2018
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018
Net Safe Bearing Capaciity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

| | |
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| Approved | PRI | PRI | PRI | Approved | PRI | PRI | PRI |

## 3. Design of Power Plant Guard Room

### 3.1 Load Calculation for Power Plant Guard Room

All the Reinforced Concrete members are designed in STAAD using ACI Code.

**L/C 3. Dead Load (DL) :**

**a) Roof Dead Load**

| | | | |
|---|---|---|---|
| i) Suspended steel channel system from ceiling = | | 0.1 | kN/m$^2$ |
| Acoustical Fiber Tile (16mm =) | | 0.05 | kN/m$^2$ |
| Sub Total = | | 0.15 | kN/m$^2$ |
| ii) 150 thk Hollow core slab = | | 2.52 | kN/m$^2$ |
| iii) Built-in Water Proofing system | | | |
| 100 mm thk Polysterene Foam | | | |
| 0.01kN/m$^2$ per 25.4mm thickness (100/25.4) = | | 0.04 | kN/m$^2$ |
| Bituminous smooth surface water proofing membrane = | | 0.07 | kN/m$^2$ |
| 90mm thick concrete screed @0.38 kN/m2 for | | | |
| 25 thickness ( = 0.38*90/25 ) | | 1.37 | kN/m$^2$ |
| 75mm thick concrete screed above HCS = 21 x0.075 = | | 1.6 | kN/m$^2$ |
| **Sub Total =** | | **5.8** | **kN/m$^2$** |

**UDL on Roof beams =** **5.8 x 2 x0.5 =** **5.8 kN/m**

**Moment due to cantilever portion of slab & screed =** (0.15*24 + (0.05+0.2)/2*20)*0.6*0.6/2 = **1.098 kNm/m**

**b) 250thk ground floor wall load =** **16x2.5x0.25 =** **10 kN/m** (Clear height of wall = 2.5m)

**L/C 4. Live Load (LR) :**

a) Roof Live Load (non-accessible) = 0.5 kN/m$^2$ (as per Civil Structural Design Criteria)

UDL on Roof beams = **0.5 x2 x0.5 =** **0.5 kN/m**

**Moment due to live load on cantilever portion of slab =** 0.5*0.6*0.6/2 = **0.09 kNm/m**

**Wind load Calculation:**

Wind load is calculated using the following parameters:

| | | | | |
|---|---|---|---|---|
| Basic Wind Speed | = | 177 | Km/hr | ( for Non-Process Plant area as per Civil, Structural Design Criteria ) |
| Exposure Category | = | D | | ( as per Civil, Structural Design Criteria ) |
| Topographic factor, $K_{zt}$ | = | 1 | | (Section 6.4.3, SBC 301 ) |
| Importance Factor, I | = | 1 | | (Section 6.5, SBC 301 ) |
| Velocity Pressure Co-efficient, $K_z$ = | | 1.04 | | (Table 7.2-2 , SBC 301 ) |
| Wind Directionality Factor, $K_d$ = | | 0.85 | | (Section 6.4.1, SBC 301 ) |

Velocity pressure, $q_z = 0.0473 \times 10^{-3} \times K_z \times K_{zt} \times K_d \times V^2 \times I$

= 1.31 kN/m$^2$

Enclosure Classificaton: Partially Enclosed Building (Section 6.2, SBC 301 )

Internal Pressure Coefficient ($GC_{pi}$) = 0.5/-0.5 (Fig 7.2-1 , SBC 301 )

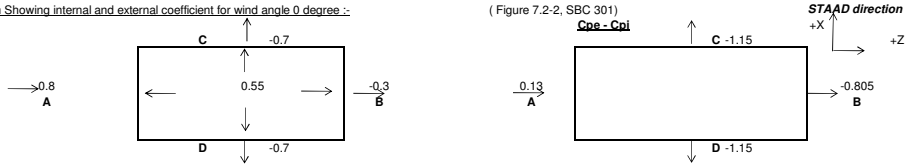
For External Pressure Coefficient ($C_{pe}$) Calculation :- (Fig 7.2-2 , SBC 301)

| | | | | | |
|---|---|---|---|---|---|
| Height of Building, h = | 4 | m | h/L = | 0.645 | |
| width of Building, B = | 4.2 | m | | | |
| Length of Building, L = | 6.2 | m | L/B = | 1.476 | |
| Max. distance between columns = | 6.2 | m | | | |

Internal pressure coefficient, $GC_{pi}$ = +/- 0.55 (Fig 7.2-1 , SBC 301)

**L/C 5. WIND IN +X DIRECTION (Cpi = +0.55) (Wind Angle = 0) (WX1)**

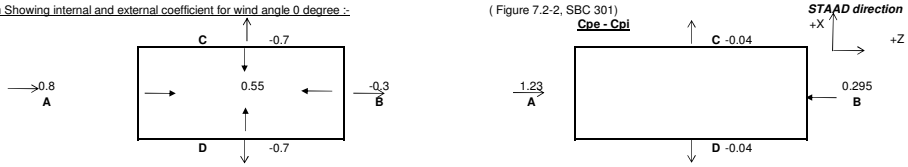
Diagram Showing internal and external coefficient for wind angle 0 degree :- ( Figure 7.2-2, SBC 301)




| | | | | |
|---|---|---|---|---|
| Area of Windward & Leeward faces each = | 3.5 x 4.2 = | 14.7 | m2 | |
| Loads on Windward face A = qz*( Gcpe-Gcpi )*A = | | 2.50 | kN | |
| Loads on Leeward face B = | | -15.50 | kN | |
| Area of other faces = | 6.2 x 3.5 = | 21.7 | m2 | |
| Load on other faces = | | -32.55 | kN | |

**L/C 6. WIND IN +X DIRECTION (Cpi = -0.55) (Wind Angle = 0) (WX2)**

Diagram Showing internal and external coefficient for wind angle 0 degree :- ( Figure 7.2-2, SBC 301)



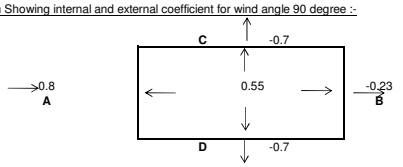

| | | | | |
|---|---|---|---|---|
| Area of Windward & Leeward faces each = | 3.5 x 4.2 = | 14.7 | m2 | |
| Loads on Windward face A = qz*( Gcpe-Gcpi )*A = | | 23.69 | kN | |
| Loads on Leeward face B = | | 5.68 | kN | |
| Area of other faces = | 6.2 x 3.5 = | 21.7 | m2 | |
| Load on other faces = | | 0.00 | kN | |

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| Approved | PRI | PRI | PRI | | | | Approved | PRI | PRI | PRI | | | |

**L/C 7. WIND IN +Z DIRECTION (Cpi = +0.55) (Wind Angle = 90) (WZ1)**

Diagram Showing internal and external coefficient for wind angle 90 degree :-



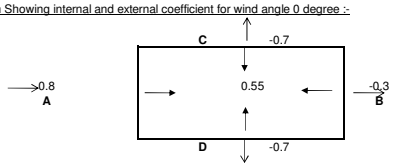

( Figure 7.2-2, SBC 301)



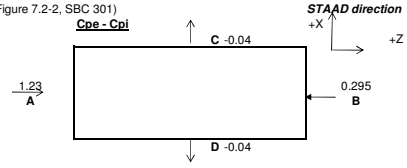

| | | | |
|---|---|---|---|
| Area of Windward & Leeward faces each = | 3.5 x 6.2 = | 21.7 | m2 |
| Loads on Windward face A = qz*( Gcpe-Gcpi )*A = | | 3.70 | kN |
| Loads on Leeward face B = | | -21.19 | kN |
| Area of other faces = | 4.2 x 3.5 = | 14.7 | m2 |
| Load on other faces = | | -22.05 | kN |

**L/C 8. WIND IN +Z DIRECTION (Cpi = -0.55) (Wind Angle = 90) (WZ2)**

Diagram Showing internal and external coefficient for wind angle 0 degree :-




( Figure 7.2-2, SBC 301)




| | | | |
|---|---|---|---|
| Area of Windward & Leeward faces each = | 3.5 x 6.2 = | 21.7 | m2 |
| Loads on Windward face A = qz*( Gcpe-Gcpi )*A = | | 34.97 | kN |
| Loads on Leeward face B = | | 8.39 | kN |
| Area of other faces = | 4.2 x 3.5 = | 14.7 | m2 |
| Load on other faces = | | 0.00 | kN |

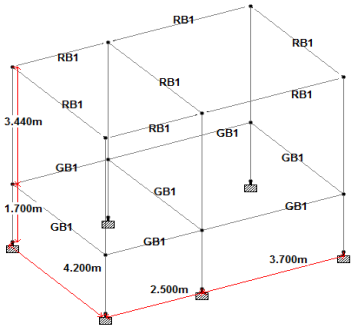
**L/C 9. Temperature Load (TL) -** Temperature Load applied for a temperature difference of $25^0$.

**L/C 1 & 2. Seismic Load (SL) - Seismic Load calculated & applied automatically in Staad as per IBC**

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| Approved | PRI | PRI | PRI | | | | Approved | PRI | PRI | PRI | | |

### 3.2 Design of Reinforced Concrete Members




3D MODEL OF GUARD ROOM REINFORCED CONCRETE FRAME IN STAAD-PRO

GRADE BEAM (GB1) - 300 WIDE X 450 DEEP

ROOF BEAM (RB1) - 300 WIDE X 450 DEEP

ALL COLUMNS ARE 350 X 350 $MM^2$ IN SIZE

All the Reinforced Concrete members except Foundation are designed in STAAD with the following factored load combinations as per SBC 304:2018

LOAD COMB 200 : (1.4 DL)

LOAD COMB 201: (1.4D+1.4T(+VE)+0.5LR)

LOAD COMB 202 : (1.4D+1.4T(-VE)+0.5LR)

LOAD COMB 203 : (1.2D+1.6WX1+0.5LR)

LOAD COMB 204 : (1.2D+1.6WX2+0.5LR)

LOAD COMB 205 : (1.2D+1.6WZ1+0.5LR)

LOAD COMB 206 : (1.2D+1.6WZ2+0.5LR)

LOAD COMB 207 : (1.2D+EZ)

LOAD COMB 208 : (1.2D+EX)

LOAD COMB 209 : (0.90D+1.6WX1)

LOAD COMB 210 : (0.90D+1.6WX2)

LOAD COMB 211: (0.90D+1.6WZ1)

LOAD COMB 212 : (0.90D+1.6WZ2)

LOAD COMB 213 : (0.90D+EZ)

LOAD COMB 214 : (0.90D+EX)

Please refer attached STAAD Output File for the following Beam & Column numbers for design

**MEMBER SCHEDULE :**

| Member mkd. | Member No. | Size | Reinforcement Details |
|---|---|---|---|
| GB1 | 25, 26, 27, 28, 29, 30, 37 | 300X450 | 3 No. - D16 both at Top & Bottom |
| RB1 | 1,2,6,7,11,13,38 | 300X450 | 3 No. - D16 both at Top & Bottom |
| COLUMNS | 3,4,5,8,9, 10, 31, 32,33,34,35,36 | 350X350 | Longitudinal Rebar 8 no. D16 |

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## 3.3 DESIGN OF FOUNDATION MKD. F1

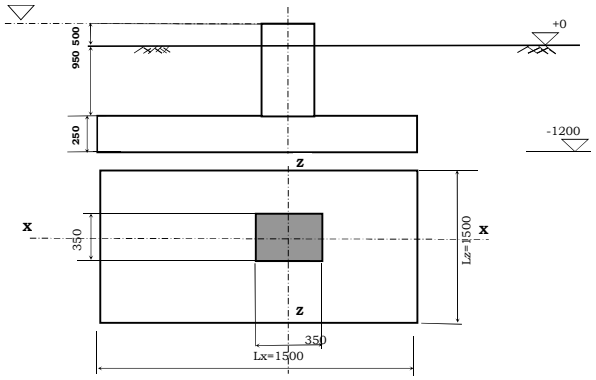
| | About-X | About-Z |
|---|---|---|
| Size of column : | 0.35m | 0.35m |
| Size of fdn. raft : | 1.50m | 1.50m |
| Thickness of raft : | 0.25 m | |
| Original ground level: | 0.00 m | |
| Finished floor level : | 0.00 m | |
| U/S level of foundation : | -1.20 m | |
| U/S level Base Plate : | 0.50 m | |
| Unit weight of conc : | 2.5 T/m$^3$ | |
| Unit weight of soil : | 1.8 T/m$^3$ | |

**SYMBOLS AND NOTATIONS**

| | |
|---|---|
| Dead Load | D |
| live load | LL |
| Roof live load | LR |
| Wind Load | W |
| Seismic Load | E |

**Calculation of Weight of Foundation :**

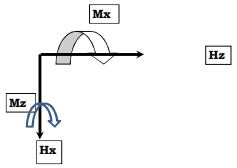
| | | |
|---|---|---|
| Weight of raft & Col.= | (1.5*1.5*0.25*2.5)+(0.35*0.35*1.45*2.5) = | 1.85T |
| Weight of soil = | ((1.5*1.5)-(0.35*0.35))*0.95*1.8 = | 3.64T |
| U/S Base Pl | Total = | 5.49T |




**Design Input:**

| | |
|---|---|
| Characteristic strength of concrete, f'$_c$= | 35 MPa |
| Yield stress of steel reinforcement, f$_y$= | 420 MPa |
| Clear cover for main reinforcement bars = | 75mm |
| Dia. of bars provd. along X direction at bottom = | 16mm |
| Dia. of bars provd. along X direction at top = | 12mm |
| Dia. of bars provided along Z direction at bottom= | 16mm |
| Dia. of bars provided along Z direction at top= | 12mm |
| Mininum percentage of reinforcement at bottom= | 0.12% |
| Mininum percentage of reinforcement at top= | 0.06% |

Effective depth of slab along X direction = 167mm = 0.167m
Effective depth of slab along Z direction = 151mm = 0.151m
Clear projection along Z direction = 0.575m
Clear projection along X direction = 0.575m




| WORKING STRESS DESIGN | Load at U/S Of Base Plt | | | | | Load at bottom of foundation | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Load Case | FX (T) | FY(T) | FZ(T) | $M_X$ | $M_Z$ | V (T) | Hx (T) | MX (T-m) | Hz (T) | MZ (T-m) |
| 100 SLS: (DL) | -0.26 | 13.95 | 0.70 | 0.37 | 0.16 | 19.44 | -0.26 | 0.37 | 0.70 | 0.16 |
| 101 SLS: (DL+T) | -0.43 | 13.22 | 1.69 | 1.39 | 0.32 | 18.71 | -0.43 | 1.39 | 1.69 | 0.32 |
| 102 SLS: (D+LR) | -0.25 | 14.22 | 0.69 | 0.37 | 0.15 | 19.71 | -0.25 | 0.37 | 0.69 | 0.15 |
| 103 SLS: (D+0.75LR) | -0.25 | 14.15 | 0.70 | 0.37 | 0.15 | 19.64 | -0.25 | 0.37 | 0.70 | 0.15 |
| 104 SLS: (D+0.75*0.7EZ) | -0.26 | 14.18 | 0.86 | 0.56 | 0.16 | 19.67 | -0.26 | 0.56 | 0.86 | 0.16 |
| 105 SLS: (D+0.75*0.7EX) | -0.05 | 14.09 | 0.70 | 0.37 | -0.04 | 19.58 | -0.05 | 0.37 | 0.70 | 0.04 |
| 106 SLS: (D+0.75WX1+0.75LR) | 0.02 | 14.30 | 0.71 | 0.38 | -0.11 | 19.79 | 0.02 | 0.38 | 0.71 | 0.11 |
| 107 SLS: (D+0.75WX2+0.75LR) | 0.03 | 14.31 | 0.70 | 0.37 | -0.11 | 19.80 | 0.03 | 0.37 | 0.70 | 0.11 |
| 108 SLS: (0.6D+WX1) | 0.22 | 8.56 | 0.43 | 0.24 | -0.25 | 14.05 | 0.22 | 0.24 | 0.43 | 0.25 |
| 109 SLS: (0.6D+WX2) | 0.23 | 8.58 | 0.42 | 0.22 | -0.26 | 14.07 | 0.23 | 0.22 | 0.42 | 0.26 |
| 110 SLS: (D+0.75WZ1+0.75LR) | -0.26 | 14.48 | 1.01 | 0.72 | 0.16 | 19.97 | -0.26 | 0.72 | 1.01 | 0.16 |
| 111 SLS: (D+0.75WZ2+0.75LR) | -0.25 | 14.51 | 1.02 | 0.73 | 0.15 | 20.00 | -0.25 | 0.73 | 1.02 | 0.15 |
| 112 SLS: (0.6D+WZ1) | -0.16 | 8.81 | 0.84 | 0.69 | 0.10 | 14.30 | -0.16 | 0.69 | 0.84 | 0.10 |
| 113 SLS: (0.6D+WZ2) | -0.15 | 8.85 | 0.86 | 0.70 | 0.09 | 14.34 | -0.15 | 0.70 | 0.86 | 0.09 |
| 114 SLS: (0.6D+0.7EX) | 0.12 | 8.55 | 0.42 | 0.22 | -0.17 | 14.04 | 0.12 | 0.22 | 0.42 | 0.17 |
| 115 SLS: (0.6D+0.7EZ) | -0.16 | 8.68 | 0.63 | 0.47 | 0.10 | 14.17 | -0.16 | 0.47 | 0.63 | 0.10 |

**CHECK FOR OVERTURNING AND SLIDING**

| | Load at Foundation Base | | | | | Overturning & Sliding Check (Coeffecient, μ = 0.4) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | V (T) | Hx (T) | Mz (Tm) | Hz (T) | Mx (Tm) | F.O.S. Mz | F.O.S. Mx | F.O.S. Hx | F.O.S. Hz |
| **Load Case** | | | | | | | | | |
| 100 SLS: (DL) | 19.44 | -0.26 | 0.16 | 0.70 | 0.37 | 94.06 | 39.19 | 29.91 | 11.11 |
| 101 SLS: (DL+T) | 18.71 | -0.43 | 0.32 | 1.69 | 1.39 | 43.58 | 10.12 | 17.40 | 4.43 |
| 102 SLS: (D+LR) | 19.71 | -0.25 | 0.15 | 0.69 | 0.37 | 95.99 | 40.17 | 31.54 | 11.43 |
| 103 SLS: (D+0.75LR) | 19.64 | -0.25 | 0.15 | 0.70 | 0.37 | 95.65 | 39.92 | 31.42 | 11.22 |
| 104 SLS: (D+0.75*0.7EZ) | 19.67 | -0.26 | 0.16 | 0.86 | 0.56 | 92.78 | 26.44 | 30.26 | 9.15 |
| 105 SLS: (D+0.75*0.7EX) | 19.58 | -0.05 | 0.04 | 0.70 | 0.37 | 349.64 | 39.48 | 156.64 | 11.19 |
| 106 SLS: (D+0.75WX1+0.75LR) | 19.79 | 0.02 | 0.11 | 0.71 | 0.38 | 141.36 | 39.16 | 395.80 | 11.15 |
| 107 SLS: (D+0.75WX2+0.75LR) | 19.80 | 0.03 | 0.11 | 0.70 | 0.37 | 133.78 | 40.24 | 264.00 | 11.31 |
| 108 SLS: (0.6D+WX1) | 14.05 | 0.22 | 0.25 | 0.43 | 0.24 | 41.82 | 44.46 | 25.55 | 13.07 |
| 109 SLS: (0.6D+WX2) | 14.07 | 0.23 | 0.26 | 0.42 | 0.22 | 40.59 | 47.32 | 24.47 | 13.40 |
| 110 SLS: (D+0.75WZ1+0.75LR) | 19.97 | -0.26 | 0.16 | 1.01 | 0.72 | 96.63 | 20.92 | 30.72 | 7.91 |
| 111 SLS: (D+0.75WZ2+0.75LR) | 20.00 | -0.25 | 0.15 | 1.02 | 0.73 | 97.40 | 20.55 | 32.00 | 7.84 |
| 112 SLS: (0.6D+WZ1) | 14.30 | -0.16 | 0.10 | 0.84 | 0.69 | 112.89 | 15.63 | 35.75 | 6.81 |
| 113 SLS: (0.6D+WZ2) | 14.34 | -0.15 | 0.09 | 0.86 | 0.70 | 115.65 | 15.28 | 38.24 | 6.67 |
| 114 SLS: (0.6D+0.7EX) | 14.04 | 0.12 | 0.17 | 0.42 | 0.22 | 61.94 | 47.22 | 46.80 | 13.37 |
| 115 SLS: (0.6D+0.7EZ) | 14.17 | -0.16 | 0.10 | 0.63 | 0.47 | 107.35 | 22.56 | 35.43 | 9.00 |

| | |
|---|---|
| Global Project ID | 900707 |
| Delivery Type | ENGINEERING PROCUREMENT CONSTRUCTION |
| Project Name | MANSOURAH MASSARAH GOLD PROJECT |
| Customer | MA'ADEN GOLD AND BASE METALS (MA'ADEN) |
| Project Country | KINGDOM OF SAUDI ARABIA |

| Ma'aden Revision No: | A01 | C01 | D01 | Outotec / L&T Revision No.: | 0 | 1 | 2 |
|---|---|---|---|---|---|---|---|
| Date | 03-07-2020 | 01-08-2020 | 26-10-2023 | Date | 03-07-2020 | 01-08-2020 | 26-10-2023 |
| Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | AS BUILT | Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | AS BUILT |
| Prepared By | SHD | SHD | SHD | Prepared By | SHD | SHD | SHD |
| Checked | SGC | SGC | SGC | Checked | SGC | SGC | SGC |
| Approved | PRI | PRI | PRI | Approved | PRI | PRI | PRI |

| ULTIMATE STRESS DESIGN | Load at U/S Of Base Plt | | | | | Load at bottom of foundation | | |
|---|---|---|---|---|---|---|---|---|
| Load Case | FX (T) | FY(T) | FZ(T) | $M_x$ | $M_z$ | V (T) | Mx (T-m) | Mz (T-m) |
| 200 ULS: (1.4 DL) | -0.36 | 19.53 | 0.98 | 0.52 | 0.22 | 25.020 | 0.52 | 0.22 |
| 201 ULS: (1.4D+1.4T(+VE)+0.5LR) | -0.61 | 18.65 | 2.36 | 1.94 | 0.45 | 24.140 | 1.94 | 0.45 |
| 202 ULS: (1.4D+1.4T(-VE)+0.5LR) | -0.11 | 20.68 | -0.40 | -0.90 | -0.02 | 26.170 | -0.90 | -0.02 |
| 203 ULS: (1.2D+1.6WX1+0.5LR) | 0.29 | 17.18 | 0.86 | 0.47 | -0.37 | 22.670 | 0.47 | -0.37 |
| 204 ULS: (1.2D+1.6WX2+0.5LR) | 0.30 | 17.21 | 0.84 | 0.44 | -0.38 | 22.700 | 0.44 | -0.38 |
| 205 ULS: (1.2D+1.6WZ1+0.5LR) | -0.31 | 17.58 | 1.51 | 1.18 | 0.19 | 23.070 | 1.18 | 0.19 |
| 206 ULS: (1.2D+1.6WZ2+0.5LR) | -0.31 | 17.64 | 1.54 | 1.21 | 0.19 | 23.130 | 1.21 | 0.19 |
| 207 ULS: (1.2D+EZ) | -0.32 | 17.18 | 1.14 | 0.80 | 0.19 | 22.670 | 0.80 | 0.19 |
| 208 ULS: (1.2D+EX) | 0.09 | 17.00 | 0.84 | 0.45 | -0.19 | 22.490 | 0.45 | -0.19 |
| 209 ULS: (0.90D+1.6WX1) | 0.36 | 12.86 | 0.65 | 0.36 | -0.41 | 18.350 | 0.36 | -0.41 |
| 210 ULS: (0.90D+1.6WX2) | 0.38 | 12.89 | 0.63 | 0.34 | -0.43 | 18.380 | 0.34 | -0.43 |
| 211 ULS: (0.90D+1.6WZ1) | -0.23 | 13.26 | 1.31 | 1.08 | 0.14 | 18.750 | 1.08 | 0.14 |
| 212 ULS: (0.90D+1.6WZ2) | -0.23 | 13.32 | 1.33 | 1.11 | 0.14 | 18.810 | 1.11 | 0.14 |
| 213 ULS: (0.90D+EZ) | -0.24 | 13.00 | 0.93 | 0.69 | 0.15 | 18.490 | 0.69 | 0.15 |
| 214 ULS: (0.90D+EX) | 0.17 | 12.82 | 0.63 | 0.34 | -0.24 | 18.310 | 0.34 | -0.24 |

**Check for Bearing Pressure**

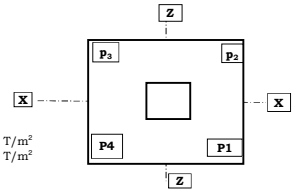
**Load Combination & Calculation of Base pressure**

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Net SBC of soil for DL, LL= 50 $T/m^2$ (BH28) Appox. Surcharge load = 2 $T/m^2$

Gross SBC of soil for DL, LL= 53.06 $T/m^2$ For WL/SL case = 62.5 $T/m^2$ For WL/SL case = 65.56 $T/m^2$




| Load Case | Design Load/moments V (T) | Mz (T-m) | Mx (T-m) | Base Pressure ($T/m^2$) V/ (Lx.Lz) | 6.Mz/ $(Lx)^2$.Lz | 6.Mx/ Lx.$(Lz)^2$ | $p_1$ ($p_{max}$) | $p_2$ | $p_4$ | $p_3$ ($p_{min}$) | ($p_{max}$-γh) | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 100 SLS: (DL) | 19.44 | 0.16 | 0.37 | 8.64 | 0.28 | 0.66 | 11.58 | 10.25 | 11.03 | 9.70 | 9.87 | O.K |
| 101 SLS: (DL+T) | 18.71 | 0.32 | 1.39 | 8.32 | 0.57 | 2.46 | 13.35 | 8.42 | 12.21 | 7.28 | 11.64 | O.K |
| 102 SLS: (D+LR) | 19.71 | 0.15 | 0.37 | 8.76 | 0.27 | 0.65 | 11.69 | 10.38 | 11.14 | 9.83 | 9.98 | O.K |
| 103 SLS: (D+0.75LR) | 19.64 | 0.15 | 0.37 | 8.73 | 0.27 | 0.66 | 11.66 | 10.35 | 11.11 | 9.80 | 9.95 | O.K |
| 104 SLS: (D+0.75*0.7EZ) | 19.67 | 0.16 | 0.56 | 8.74 | 0.28 | 0.99 | 12.02 | 10.03 | 11.45 | 9.47 | 10.31 | O.K |
| 105 SLS: (D+0.75*0.7EX) | 19.58 | 0.04 | 0.37 | 8.70 | 0.07 | 0.66 | 11.44 | 10.12 | 11.29 | 9.97 | 9.73 | O.K |
| 106 SLS: (D+0.75WX1+0.75LR) | 19.79 | 0.11 | 0.38 | 8.80 | 0.19 | 0.67 | 11.66 | 10.31 | 11.28 | 9.94 | 9.95 | O.K |
| 107 SLS: (D+0.75WX2+0.75LR) | 19.80 | 0.11 | 0.37 | 8.80 | 0.20 | 0.66 | 11.65 | 10.34 | 11.26 | 9.95 | 9.94 | O.K |
| 108 SLS: (0.6D+WX1) | 14.05 | 0.25 | 0.24 | 6.24 | 0.45 | 0.42 | 9.11 | 8.27 | 8.22 | 7.38 | 7.40 | O.K |
| 109 SLS: (0.6D+WX2) | 14.07 | 0.26 | 0.22 | 6.25 | 0.46 | 0.40 | 9.11 | 8.32 | 8.19 | 7.39 | 7.40 | O.K |
| 110 SLS: (D+0.75WZ1+0.75LR) | 19.97 | 0.16 | 0.72 | 8.88 | 0.28 | 1.27 | 12.42 | 9.88 | 11.87 | 9.33 | 10.71 | O.K |
| 111 SLS: (D+0.75WZ2+0.75LR) | 20.00 | 0.15 | 0.73 | 8.89 | 0.27 | 1.30 | 12.46 | 9.86 | 11.91 | 9.32 | 10.75 | O.K |
| 112 SLS: (0.6D+WZ1) | 14.30 | 0.10 | 0.69 | 6.36 | 0.17 | 1.22 | 9.74 | 7.30 | 9.41 | 6.97 | 8.03 | O.K |
| 113 SLS: (0.6D+WZ2) | 14.34 | 0.09 | 0.70 | 6.37 | 0.17 | 1.25 | 9.79 | 7.29 | 9.46 | 6.96 | 8.08 | O.K |
| 114 SLS: (0.6D+0.7EX) | 14.04 | 0.17 | 0.22 | 6.24 | 0.30 | 0.40 | 8.94 | 8.15 | 8.33 | 7.54 | 7.23 | O.K |
| 115 SLS: (0.6D+0.7EZ) | 14.17 | 0.10 | 0.47 | 6.30 | 0.18 | 0.84 | 9.31 | 7.64 | 8.96 | 7.28 | 7.60 | O.K |

***Calculation of Design Pressure***

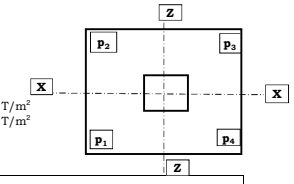
**Load Combination & Calculation of Base pressure**

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Net SBC of soil for DL, LL= 50 $T/m^2$ Surcharge load = 2 $T/m^2$

Gross SBC of soil for DL, LL= 53.06 $T/m^2$ For WL/SL case = 62.5 $T/m^2$ For WL/SL case = 65.56 $T/m^2$




| Load Case | Design Load/moments V (T) | Mz (T-m) | Mx (T-m) | V/ (Lx.Lz) | 6.Mz/ $(Lx)^2$.Lz | 6.Mx/ Lx.$(Lz)^2$ | $p_1$ ($p_{max}$) | $p_2$ | $p_4$ | $p_3$ ($p_{min}$) | ($p_{max}$-γh) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 200 ULS: (1.4 DL) | 25.0 | 0.22 | 0.52 | 11.12 | 0.39 | 0.93 | 14.43 | 12.58 | 13.66 | 11.81 | 12.27 |
| 201 ULS: (1.4D+1.4T(+VE)+0.5LR | 24.1 | 0.45 | 1.94 | 10.73 | 0.80 | 3.45 | 16.98 | 8.08 | 13.38 | 6.48 | 14.82 |
| 202 ULS: (1.4D+1.4T(-VE)+0.5LR) | 26.2 | -0.02 | -0.90 | 11.63 | 0.03 | 1.60 | 15.26 | 10.06 | 13.20 | 10.00 | 13.10 |
| 203 ULS: (1.2D+1.6WX1+0.5LR) | 22.7 | -0.37 | 0.47 | 10.08 | 0.65 | 0.83 | 13.55 | 9.90 | 10.25 | 8.60 | 11.39 |
| 204 ULS: (1.2D+1.6WX2+0.5LR) | 22.7 | -0.38 | 0.44 | 10.09 | 0.68 | 0.79 | 13.55 | 9.98 | 10.20 | 8.62 | 11.39 |
| 205 ULS: (1.2D+1.6WZ1+0.5LR) | 23.1 | 0.19 | 1.18 | 10.25 | 0.34 | 2.10 | 14.69 | 8.48 | 12.02 | 7.81 | 12.53 |
| 206 ULS: (1.2D+1.6WZ2+0.5LR) | 23.1 | 0.19 | 1.21 | 10.28 | 0.33 | 2.16 | 14.77 | 8.45 | 12.11 | 7.79 | 12.61 |
| 207 ULS: (1.2D+EZ) | 22.7 | 0.19 | 0.80 | 10.08 | 0.34 | 1.42 | 13.84 | 9.00 | 11.15 | 8.31 | 11.68 |
| 208 ULS: (1.2D+EX) | 22.5 | -0.19 | 0.45 | 10.00 | 0.34 | 0.79 | 13.13 | 9.54 | 10.45 | 8.86 | 10.97 |
| 209 ULS: (0.90D+1.6WX1) | 18.4 | -0.41 | 0.36 | 8.16 | 0.73 | 0.63 | 11.52 | 8.26 | 8.06 | 6.79 | 9.36 |
| 210 ULS: (0.90D+1.6WX2) | 18.4 | -0.43 | 0.34 | 8.17 | 0.76 | 0.60 | 11.52 | 8.33 | 8.01 | 6.82 | 9.36 |
| 211 ULS: (0.90D+1.6WZ1) | 18.8 | 0.14 | 1.08 | 8.33 | 0.25 | 1.91 | 12.50 | 6.68 | 9.99 | 6.17 | 10.34 |
| 212 ULS: (0.90D+1.6WZ2) | 18.8 | 0.14 | 1.11 | 8.36 | 0.25 | 1.96 | 12.57 | 6.64 | 10.08 | 6.15 | 10.41 |
| 213 ULS: (0.90D+EZ) | 18.5 | 0.15 | 0.69 | 8.22 | 0.26 | 1.22 | 11.70 | 7.25 | 9.18 | 6.73 | 9.54 |
| 214 ULS: (0.90D+EX) | 18.3 | -0.24 | 0.34 | 8.14 | 0.42 | 0.60 | 11.15 | 7.96 | 8.31 | 7.12 | 8.99 |

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|---|---|
| Global Project ID | 900707 |
| Delivery Type | ENGINEERING PROCUREMENT CONSTRUCTION |
| Project Name | MANSOURAH MASSARAH GOLD PROJECT |
| Customer | MA'ADEN GOLD AND BASE METALS (MA'ADEN) |
| Project Country | KINGDOM OF SAUDI ARABIA |

| Ma'aden Revision No: | A01 | C01 | D01 |
|---|---|---|---|
| Date | 03-07-2020 | 01-08-2020 | 26-10-2023 |
| Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | AS BUILT |
| Prepared By | SHD | SHD | SHD |
| Checked | SGC | SGC | SGC |
| Approved | PRI | PRI | PRI |

| Outotec / L&T Revision No.: | 0 | 1 | 2 |
|---|---|---|---|
| Date | 03-07-2020 | 01-08-2020 | 26-10-2023 |
| Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | AS BUILT |
| Prepared By | SHD | SHD | SHD |
| Checked | SGC | SGC | SGC |
| Approved | PRI | PRI | PRI |

**Calculation of Bending Moment and Shear Force in X direction :**

$BM_{1x}$ =Moment at the face of column at section 1x-1x in T-m/m.

$BM_{1z}$ =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.35+2*0.5*0.151)+(0.35+2*0.5*0.151))*2

= 2.004 m

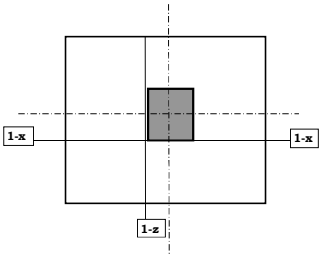
Punching Area= ((0.35+2*0.5*0.151)*(0.35+2*0.5*0.151))

= 0.25 $m^2$ Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.25*2.5+0.95*1.8)

= 2.34 $T/m^2$

Area of footing, $A_f$= 2.25 $m^2$




**Check against Two-way shear force at a distance d/2 from face of column:**

Design punching shear at a distance d/2 from face of column, Vu= 29.64 T

Strength of Concrete in shear Vc1 = 64.72 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 74.80 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 59.67 T **Table 22.6.5.2_ (c)**

Design Vc= 59.67 T

Allowable shear stress, ($\phi$ Vc)/bd = 1.48 $N/mm^2$

Design punching shear stress, $\tau$ = (29.64*10000)/(2004*151)

= 0.98 $N/mm^2$ **No shear reinforcement reqd.**

| Load Case | $p_{1x}$ ($T/m^2$) | $B.M_{1x}$ ™ | $S.F_{(oneway)}$ (T) | $B.M_{1z}$ | $S.F_{(Oneway)}$ (T) | $S.F_{(punching)}$ | $V_u/b_0d$ | Punching check (Remarks) | Allowable shear ($0.75*\sqrt{fc}/6$ ($N/mm^2$) | Actual Shear | Remarks | Mu/bd2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ultimate Conditions | | | | | | | | | | | | |
| 200 ULS: (1.4 DL) | 12.27 | 2.03 | 5.01 | 2.03 | 5.20 | 24.54 | 0.81 | OK | 0.74 | 0.345 | OK | 0.89 |
| 201 ULS: (1.4D+1.4T(+VE)+0.5LR | 14.82 | 2.45 | 6.05 | 2.45 | 6.28 | 29.64 | 0.98 | OK | 0.74 | 0.416 | OK | 1.07 |
| 202 ULS: (1.4D+1.4T(-VE)+0.5LR) | 13.10 | 2.17 | 5.34 | 2.17 | 5.55 | 26.20 | 0.87 | OK | 0.74 | 0.368 | OK | 0.95 |
| 203 ULS: (1.2D+1.6WX1+0.5LR) | 11.39 | 1.88 | 4.65 | 1.88 | 4.83 | 22.78 | 0.75 | OK | 0.74 | 0.32 | OK | 0.83 |
| 204 ULS: (1.2D+1.6WX2+0.5LR) | 11.39 | 1.88 | 4.65 | 1.88 | 4.83 | 22.78 | 0.75 | OK | 0.74 | 0.32 | OK | 0.83 |
| 205 ULS: (1.2D+1.6WZ1+0.5LR) | 12.53 | 2.07 | 5.11 | 2.07 | 5.31 | 25.06 | 0.83 | OK | 0.74 | 0.352 | OK | 0.91 |
| 206 ULS: (1.2D+1.6WZ2+0.5LR) | 12.61 | 2.08 | 5.14 | 2.08 | 5.35 | 25.22 | 0.83 | OK | 0.74 | 0.354 | OK | 0.91 |
| 207 ULS: (1.2D+EZ) | 11.68 | 1.93 | 4.77 | 1.93 | 4.95 | 23.36 | 0.77 | OK | 0.74 | 0.328 | OK | 0.85 |
| 208 ULS: (1.2D+EX) | 10.97 | 1.81 | 4.48 | 1.81 | 4.65 | 21.94 | 0.73 | OK | 0.74 | 0.308 | OK | 0.80 |
| 209 ULS: (0.90D+1.6WX1) | 9.36 | 1.55 | 3.82 | 1.55 | 3.97 | 18.72 | 0.62 | OK | 0.74 | 0.263 | OK | 0.68 |
| 210 ULS: (0.90D+1.6WX2) | 9.36 | 1.55 | 3.82 | 1.55 | 3.97 | 18.72 | 0.62 | OK | 0.74 | 0.263 | OK | 0.68 |
| 211 ULS: (0.90D+1.6WZ1) | 10.34 | 1.71 | 4.22 | 1.71 | 4.38 | 20.68 | 0.68 | OK | 0.74 | 0.29 | OK | 0.75 |
| 212 ULS: (0.90D+1.6WZ2) | 10.41 | 1.72 | 4.25 | 1.72 | 4.41 | 20.82 | 0.69 | OK | 0.74 | 0.292 | OK | 0.75 |
| 213 ULS: (0.90D+EZ) | 9.54 | 1.58 | 3.89 | 1.58 | 4.04 | 19.08 | 0.63 | OK | 0.74 | 0.268 | OK | 0.69 |
| 214 ULS: (0.90D+EX) | 8.99 | 1.49 | 3.67 | 1.49 | 3.81 | 17.98 | 0.59 | OK | 0.74 | 0.252 | OK | 0.65 |

| Load Case | kn (Ø=0.75) | p = | $A_{st}$ |
|---|---|---|---|
| Ultimate Conditions | | | |
| 200 ULS: (1.4 DL) | 1.19 | 0.003 | 501 |
| 201 ULS: (1.4D+1.4T(+VE)+0.5LR | 1.43 | 0.004 | 668 |
| 202 ULS: (1.4D+1.4T(-VE)+0.5LR) | 1.27 | 0.003 | 501 |
| 203 ULS: (1.2D+1.6WX1+0.5LR) | 1.1 | 0.003 | 501 |
| 204 ULS: (1.2D+1.6WX2+0.5LR) | 1.1 | 0.003 | 501 |
| 205 ULS: (1.2D+1.6WZ1+0.5LR) | 1.21 | 0.003 | 501 |
| 206 ULS: (1.2D+1.6WZ2+0.5LR) | 1.22 | 0.003 | 501 |
| 207 ULS: (1.2D+EZ) | 1.13 | 0.003 | 501 |
| 208 ULS: (1.2D+EX) | 1.06 | 0.003 | 501 |
| 209 ULS: (0.90D+1.6WX1) | 0.9 | 0.002 | 334 |
| 210 ULS: (0.90D+1.6WX2) | 0.9 | 0.002 | 334 |
| 211 ULS: (0.90D+1.6WZ1) | 1 | 0.002 | 334 |
| 212 ULS: (0.90D+1.6WZ2) | 1.01 | 0.002 | 334 |
| 213 ULS: (0.90D+EZ) | 0.92 | 0.002 | 334 |
| 214 ULS: (0.90D+EX) | 0.87 | 0.002 | 334 |

| Location | Mux (T-m) | $\phi$ Kn = $M_{u1}/bd^2$ | Kn ($N/mm^2$) | ρ= | Ast ($mm^2$) | Minm. Ast ($mm^2$) | Bar Dia | Spacing reqd.(mm) | Maxm spacing | Spacing provd. |
|---|---|---|---|---|---|---|---|---|---|---|
| At bottom | 2.45 | 1.07 | 1.43 | 0.004 | 668.00 | 300.60 | 12ϕT | 169.31 | 300 | 150 |
| At top | 0.00 | 0.00 | 0.00 | 0.0018 | 271.80 | 271.80 | 10ϕT | 288.96 | 300 | 150 |

**SUMMARY OF REINFORCEMENT IN RAFT:**

Along X Direction : At Bottom **12ϕT** **150** c/c At Top **10ϕT** **150** c/c

Along Z Direction : At Bottom **12ϕT** **150** c/c At Top **10ϕT** **150** c/c

| | |
|---|---|
| Global Project ID | 900707 |
| Delivery Type | ENGINEERING PROCUREMENT CONSTRUCTION |
| Project Name | MANSOURAH MASSARAH GOLD PROJECT |
| Customer | MA'ADEN GOLD AND BASE METALS (MA'ADEN) |
| Project Country | KINGDOM OF SAUDI ARABIA |

| Ma'aden Revision No: | A01 | C01 | D01 |
|---|---|---|---|
| Date | 03-07-2020 | 01-08-2020 | 26-10-2023 |
| Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | AS BUILT |
| Prepared By | SHD | SHD | SHD |
| Checked | SGC | SGC | SGC |
| Approved | PRI | PRI | PRI |

| Outotec / L&T Revision No.: | 0 | 1 | 2 |
|---|---|---|---|
| Date | 03-07-2020 | 01-08-2020 | 26-10-2023 |
| Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | AS BUILT |
| Prepared By | SHD | SHD | SHD |
| Checked | SGC | SGC | SGC |
| Approved | PRI | PRI | PRI |

## REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

| | |
|---|---|
| Global Project ID | **900707** |
| Delivery Type | **ENGINEERING PROCUREMENT CONSTRUCTION** |
| Project Name | **MANSOURAH MASSARAH GOLD PROJECT** |
| Customer | **MA'ADEN GOLD AND BASE METALS (MA'ADEN)** |
| Project Country | **KINGDOM OF SAUDI ARABIA** |

| **Ma'aden Revision No:** | A01 | C01 | D01 | | | | Outotec / L&T Revision No.: | 0 | 1 | 2 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Date | 03-07-2020 | 01-08-2020 | 26-10-2023 | | | | Date | 03-07-2020 | 01-08-2020 | 26-10-2023 | | |
| Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | AS BUILT | | | | Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | AS BUILT | | |
| Prepared By | SHD | SHD | SHD | | | | Prepared By | SHD | SHD | SHD | | |
| Checked | SGC | SGC | SGC | | | | Checked | SGC | SGC | SGC | | |
| Approved | PRI | PRI | PRI | | | | Approved | PRI | PRI | PRI | | |

**ATTACHMENT 1 STAAD Input & Output File of Guard Room**

```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    JUL  3, 2020                   *
*           Time=    13:12:24                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: Guard room.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 18-OCT-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 2.5 0 0; 3 6.2 0 0; 4 0 3.43961 0; 5 2.5 3.44 0; 6 6.2 3.43961 0
     9. 7 0 0 4.2; 8 2.5 0 4.2; 9 6.2 0 4.2; 10 0 3.43961 4.2; 11 2.5 3.44 4.2
    10. 12 6.2 3.44 4.2; 23 0 -1.7 0; 24 2.5 -1.7 0; 25 6.2 -1.7 0; 26 0 -1.7 4.2
    11. 27 2.5 -1.7 4.2; 28 6.2 -1.7 4.2
    12. MEMBER INCIDENCES
    13. 1 4 5; 2 5 6; 3 1 4; 4 2 5; 5 3 6; 6 10 11; 7 11 12; 8 7 10; 9 8 11; 10 9 12
    14. 11 4 10; 13 6 12; 25 1 2; 26 2 3; 27 3 9; 28 9 8; 29 8 7; 30 7 1; 31 1 23
    15. 32 2 24; 33 3 25; 34 7 26; 35 8 27; 36 9 28; 37 2 8; 38 5 11
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17184E+007
    19. POISSON 0.17
    20. DENSITY 23.6158
    21. ALPHA 5E-006
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27578.9
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 3 5 8 10 31 33 34 36 PRIS YD 0.35 ZD 0.35
    28. 1 2 6 7 PRIS YD 0.45 ZD 0.3
    29. MEMBER PROPERTY AMERICAN
    30. 27 30 37 PRIS YD 0.45 ZD 0.3
    31. MEMBER PROPERTY AMERICAN
    32. 11 13 38 PRIS YD 0.45 ZD 0.3
    33. MEMBER PROPERTY AMERICAN
    34. 4 9 32 35 PRIS YD 0.35 ZD 0.35
    35. MEMBER PROPERTY AMERICAN
    36. 25 26 28 29 PRIS YD 0.45 ZD 0.3
    37. CONSTANTS
    38. BETA 90 MEMB 4 9 32 35
```

```
39. MATERIAL CONCRETE ALL
40. SUPPORTS
41. 23 TO 28 FIXED
42. ******************************************************************************
43. MEMBER OFFSET
44. 2 7 START 0.2 0 0
45. 2 7 END -0.125 0 0
46. 1 6 START 0.125 0 0
47. 1 6 END -0.2 0 0
48. 11 13 START 0 0 0.2
49. 11 13 END 0 0 -0.2
50. DEFINE IBC 2003
51. SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
52. SELFWEIGHT 1
53. MEMBER WEIGHT
54. 27 30 UNI 10
55. MEMBER WEIGHT
56. 1 2 6 7 UNI 8.9
57. DEFINE WIND LOAD
```

***** NOTE: If any floor diaphragm is present in the model Wind Load definition should be defined after Floor Diaphragm definition. Otherwise wind load generation may be unsuccessful during analysis.**

```
58. TYPE 1 WIND X WINDWARD
59. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
60. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
61. 1 4.000 M 14.000 M 5.000 M 2.000 0.010 0 -
62. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
63. 0 0 0 0.900 0.800 -0.550
64. !> END GENERATED DATA BLOCK
65. INT 1.6486 1.6486 HEIG  0  4.572
66. TYPE 2 WIND X LEEWARD
67. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
68. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
69. 1 4.000 M 14.000 M 5.000 M 2.000 0.010 1 -
70. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
71. 0 0 0 0.900 -0.200 0.550
72. !> END GENERATED DATA BLOCK
73. INT -0.947728 -0.947728 HEIG  0  4
74. TYPE 3 SIDEWARD X
75. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
76. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
77. 1 4.000 M 15.000 M 5.000 M 2.000 0.010 2 -
78. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
79. 0 0 0 0.900 -0.700 0.550
80. !> END GENERATED DATA BLOCK
81. INT -1.53179 -1.53179 HEIG  0  4
82. TYPE 4 WIND Z WINDWARD
83. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
84. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
85. 1 4.000 M 5.000 M 15.000 M 2.000 0.010 0 -
86. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
87. 0 0 0 0.886 0.800 -0.550
88. !> END GENERATED DATA BLOCK
89. INT 1.63434 1.63434 HEIG  0  4.572
```

```
 90. TYPE 5 WIND Z LEEWARD
 91. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
 92. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
 93. 1 4.000 M 5.000 M 15.000 M 2.000 0.010 1 -
 94. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
 95. 0 0 0 0.886 -0.500 0.550
 96. !> END GENERATED DATA BLOCK
 97. INT -1.28925 -1.28925 HEIG  0  4
 98. TYPE 6 SIDEWARD Z
 99. <! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
100. ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
101. 1 4.000 M 5.000 M 15.000 M 2.000 0.010 2 -
102. 0 0 0 0 1.030 1.000 1.000 0.850 0 -
103. 0 0 0 0.886 -0.700 0.550
104. !> END GENERATED DATA BLOCK
105. INT -1.51931 -1.51931 HEIG  0  4
106. LOAD 1 LOADTYPE SEISMIC  TITLE EQK LOAD Z
107. IBC LOAD Z 1
108. PERFORM ANALYSIS
```

**P R O B L E M  S T A T I S T I C S**

**---------------------------------**

| | | | |
|---|---|---|---|
| **NUMBER OF JOINTS** | **18** | **NUMBER OF MEMBERS** | **26** |
| **NUMBER OF PLATES** | **0** | **NUMBER OF SOLIDS** | **0** |
| **NUMBER OF SURFACES** | **0** | **NUMBER OF SUPPORTS** | **6** |

**SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER**

**ORIGINAL/FINAL BAND-WIDTH= 12/ 7/ 36 DOF**
**TOTAL PRIMARY LOAD CASES = 1, TOTAL DEGREES OF FREEDOM = 72**
**TOTAL LOAD COMBINATION CASES = 0 SO FAR.**
**SIZE OF STIFFNESS MATRIX = 3 DOUBLE KILO-WORDS**
**REQRD/AVAIL. DISK SPACE = 12.1/ 111788.3 MB**

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.016 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.203 T =  0.189 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.189                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.042 X       424.72   *
*                       =        18.01  KN                 *
************************************************************
```

```
109. CHANGE
110. LOAD 2 LOADTYPE SEISMIC  TITLE EQK LOAD X
111. IBC LOAD X 1
112. PERFORM ANALYSIS
```

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.016 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.203 T =  0.172 Tuser = 0.000                 *
*  TIME PERIOD USED (T) =  0.172                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.042 X       424.72   *
*                       =        18.01  KN                 *
************************************************************
```

```
113. CHANGE
114. LOAD 3 LOADTYPE DEAD  TITLE DL
115. SELFWEIGHT Y -1
116. MEMBER LOAD
117. 25 TO 30 37 UNI GY -10
118. MEMBER LOAD
119. 1 2 6 7 11 13 38 UNI GY -5.8
120. MEMBER LOAD
121. 1 2 6 7 11 13 UMOM X 1
122. *11 UMOM GZ 1
123. *13 UMOM GZ -1
124. *1 2 UMOM GX -1
125. *6 7 UMOM GX 1
126. LOAD 4 LOADTYPE LIVE  TITLE LL
```

```
127. MEMBER LOAD
128. 1 2 6 7 11 13 38 UNI GY -0.5
129. 1 2 6 7 11 13 UMOM X 0.09
130. *11 UMOM GZ 0.1
131. *13 UMOM GZ -0.1
132. *1 2 UMOM GX -0.1
133. *6 7 UMOM GX 0.1
134. LOAD 5 LOADTYPE WIND  TITLE WX1
135. JOINT LOAD
136. 1 4 7 10 FX 0.625
137. 3 6 9 12 FX 3.875
138. 1 TO 6 FZ -5.425
139. 7 TO 12 FZ 5.425
140. LOAD 6 LOADTYPE WIND  TITLE WX2
141. JOINT LOAD
142. 1 4 7 10 FX 6
143. 3 6 9 12 FX -1.42
144. LOAD 7 LOADTYPE WIND  TITLE WZ1
145. JOINT LOAD
146. 1 TO 6 FZ 0.62
147. 7 TO 12 FZ 3.5
148. 1 4 7 10 FX -5.5
149. 3 6 9 12 FX 5.5
150. LOAD 8 LOADTYPE WIND  TITLE WZ2
151. JOINT LOAD
152. 1 TO 6 FZ 5.8
153. 7 TO 12 FZ -1.4
154. LOAD 9 LOADTYPE TEMPERATURE  TITLE TEMP
155. TEMPERATURE LOAD
156. 1 TO 11 13 25 TO 30 37 38 TEMP 25
157. ************************************************
158. LOAD COMB 100 SLS: (DL)
159. 3 1.0
160. LOAD COMB 101 SLS: (DL+T)
161. 3 1.0 9 1.0
162. LOAD COMB 102 SLS: (D+LR)
163. 3 1.0 4 1.0
164. LOAD COMB 103 SLS: (D+0.75LR)
165. 3 1.0 4 0.75
166. LOAD COMB 104 SLS: (D+0.75*0.7EZ)
167. 3 1.0 2 0.525
168. LOAD COMB 105 SLS: (D+0.75*0.7EX)
169. 3 1.0 1 0.525
170. LOAD COMB 106 SLS: (D+0.75WX1+0.75LR)
171. 3 1.0 5 0.75 4 0.75
172. LOAD COMB 107 SLS: (D+0.75WX2+0.75LR)
173. 3 1.0 6 0.75 4 0.75
174. LOAD COMB 108 SLS: (0.6D+WX1)
175. 3 0.6 5 1.0
176. LOAD COMB 109 SLS: (0.6D+WX2)
177. 3 0.6 6 1.0
178. ******************************************
179. LOAD COMB 110 SLS: (D+0.75WZ1+0.75LR)
180. 3 1.0 7 0.75 4 0.75
181. LOAD COMB 111 SLS: (D+0.75WZ2+0.75LR)
182. 3 1.0 8 0.75 4 0.75
```

```
183. LOAD COMB 112 SLS: (0.6D+WZ1)
184. 3 0.6 7 1.0
185. LOAD COMB 113 SLS: (0.6D+WZ2)
186. 3 0.6 8 1.0
187. ********************************************
188. LOAD COMB 114 SLS: (0.6D+0.7EX)
189. 3 0.6 2 0.7
190. LOAD COMB 115 SLS: (0.6D+0.7EZ)
191. 3 0.6 1 0.7
192. ************************************************
193. ******************************************************
194. LOAD COMB 200 ULS: (1.4 DL)
195. 3 1.4
196. LOAD COMB 201 ULS: (1.4D+1.4T(+VE)+0.5LR)
197. 3 1.4 4 0.5 9 1.4
198. LOAD COMB 202 ULS: (1.4D+1.4T(-VE)+0.5LR)
199. 3 1.4 4 0.5 9 -1.4
200. LOAD COMB 203 ULS: (1.2D+1.6WX1+0.5LR)
201. 3 1.2 5 1.6 4 0.5
202. LOAD COMB 204 ULS: (1.2D+1.6WX2+0.5LR)
203. 3 1.2 6 1.6 4 0.5
204. LOAD COMB 205 ULS: (1.2D+1.6WZ1+0.5LR)
205. 3 1.2 4 0.5 7 1.6
206. LOAD COMB 206 ULS: (1.2D+1.6WZ2+0.5LR)
207. 3 1.2 4 0.5 8 1.6
208. LOAD COMB 207 ULS: (1.2D+EZ)
209. 3 1.2 1 1.0
210. LOAD COMB 208 ULS: (1.2D+EX)
211. 3 1.2 2 1.0
212. LOAD COMB 209 ULS: (0.90D+1.6WX1)
213. 3 0.9 5 1.6
214. LOAD COMB 210 ULS: (0.90D+1.6WX2)
215. 3 0.9 6 1.6
216. LOAD COMB 211 ULS: (0.90D+1.6WZ1)
217. 3 0.9 7 1.6
218. LOAD COMB 212 ULS: (0.90D+1.6WZ2)
219. 3 0.9 8 1.6
220. LOAD COMB 213 ULS: (0.90D+EZ)
221. 3 0.9 1 1.0
222. LOAD COMB 214 ULS: (0.90D+EX)
223. 3 0.9 2 1.0
224. ************************************************************
225. PERFORM ANALYSIS PRINT STATICS CHECK
```

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE DEAD TITLE DL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.297673386E+01
Y = 0.125750852E+01
Z = 0.211777109E+01

***TOTAL APPLIED LOAD ( KN METE ) SUMMARY (LOADING 3 )
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -624.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1323.07 MY= 0.00 MZ= -1859.70

***TOTAL REACTION LOAD( KN METE ) SUMMARY (LOADING 3 )
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 624.75
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1323.07 MY= 0.00 MZ= 1859.70

MAXIMUM DISPLACEMENTS ( CM /RADIANS) (LOADING 3)

| | MAXIMUMS | AT NODE |
|---|---|---|
| X = | -8.72352E-03 | 12 |
| Y = | -1.53688E-02 | 11 |
| Z = | 4.91195E-03 | 5 |
| RX= | 2.36547E-04 | 5 |
| RY= | -1.78009E-06 | 4 |
| RZ= | 1.59534E-04 | 12 |

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE LIVE TITLE LL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.295117895E+01
Y = 0.343983725E+01
Z = 0.218724898E+01

```
***TOTAL APPLIED LOAD ( KN   METE ) SUMMARY (LOADING     4 )
    SUMMATION FORCE-X =           0.00
    SUMMATION FORCE-Y =         -11.45
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=          25.04  MY=           0.00  MZ=         -33.79


***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING     4 )
    SUMMATION FORCE-X =           0.00
    SUMMATION FORCE-Y =          11.45
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=         -25.04  MY=           0.00  MZ=          33.79


MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING      4)
          MAXIMUMS    AT NODE
   X = -3.87533E-04      12
   Y = -5.35871E-04      11
   Z =  4.23446E-04       5
   RX=  1.70532E-05       5
   RY= -1.54356E-07       4
   RZ=  1.07623E-05      12


       STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.     5
       LOADTYPE WIND  TITLE WX1


        CENTER OF FORCE BASED ON X FORCES ONLY (METE).
      (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                    X =  0.533888864E+01
                    Y =  0.171988902E+01
                    Z =  0.210000009E+01

***TOTAL APPLIED LOAD ( KN   METE ) SUMMARY (LOADING     5 )
    SUMMATION FORCE-X =          18.00
    SUMMATION FORCE-Y =           0.00
    SUMMATION FORCE-Z =           0.00

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=           0.00  MY=          37.80  MZ=         -30.96


***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING     5 )
    SUMMATION FORCE-X =         -18.00
    SUMMATION FORCE-Y =           0.00
    SUMMATION FORCE-Z =           0.00
```

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -37.80 MZ= 30.96

MAXIMUM DISPLACEMENTS ( CM /RADIANS) (LOADING 5)

| | MAXIMUMS | AT NODE |
|---|---|---|
| X = | 3.85638E-02 | 12 |
| Y = | 5.17748E-04 | 10 |
| Z = | -3.89511E-04 | 5 |
| RX= | -1.75453E-06 | 2 |
| RY= | 1.54094E-07 | 4 |
| RZ= | -5.12921E-05 | 3 |

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
LOADTYPE WIND TITLE WX2

***TOTAL APPLIED LOAD ( KN METE ) SUMMARY (LOADING 6 )
SUMMATION FORCE-X = 18.32
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 38.47 MZ= -31.51

***TOTAL REACTION LOAD( KN METE ) SUMMARY (LOADING 6 )
SUMMATION FORCE-X = -18.32
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -38.47 MZ= 31.51

MAXIMUM DISPLACEMENTS ( CM /RADIANS) (LOADING 6)

| | MAXIMUMS | AT NODE |
|---|---|---|
| X = | 3.93111E-02 | 10 |
| Y = | 5.28008E-04 | 10 |
| Z = | 4.23493E-07 | 10 |
| RX= | 4.45621E-10 | 1 |
| RY= | 2.74605E-09 | 10 |
| RZ= | -5.00297E-05 | 3 |

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
LOADTYPE WIND TITLE WZ1

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.289999995E+01
Y = 0.171992527E+01
Z = 0.356796133E+01

***TOTAL APPLIED LOAD ( KN METE ) SUMMARY (LOADING 7 )
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 24.72

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 42.52 MY= -71.69 MZ= 0.00

***TOTAL REACTION LOAD( KN METE ) SUMMARY (LOADING 7 )
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -24.72

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -42.52 MY= 71.69 MZ= 0.00

MAXIMUM DISPLACEMENTS ( CM /RADIANS) (LOADING 7)

| | MAXIMUMS | AT NODE |
|---|---|---|
| X = | 6.07912E-04 | 9 |
| Y = | -5.31314E-04 | 10 |
| Z = | 6.29035E-02 | 11 |
| RX= | 8.21811E-05 | 7 |
| RY= | 6.84574E-07 | 12 |
| RZ= | -2.24622E-06 | 9 |

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
LOADTYPE WIND TITLE WZ2

***TOTAL APPLIED LOAD ( KN METE ) SUMMARY (LOADING 8 )
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 26.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 45.40 MY= -76.56 MZ= 0.00

***TOTAL REACTION LOAD( KN METE ) SUMMARY (LOADING 8 )
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -26.40

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -45.40 MY= 76.56 MZ= 0.00

MAXIMUM DISPLACEMENTS ( CM /RADIANS) (LOADING 8)

| | MAXIMUMS | AT NODE |
|---|---|---|
| X = | 4.88844E-06 | 12 |
| Y = | 5.66266E-04 | 4 |
| Z = | 6.73265E-02 | 5 |
| RX= | 8.84208E-05 | 1 |
| RY= | 7.70195E-07 | 6 |
| RZ= | 1.54195E-07 | 10 |

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
LOADTYPE TEMPERATURE TITLE TEMP

***TOTAL APPLIED LOAD ( KN METE ) SUMMARY (LOADING 9 )
SUMMATION FORCE-X = 0.0000000E+00
SUMMATION FORCE-Y = -6.3213032E-14
SUMMATION FORCE-Z = 0.0000000E+00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4.6230848E-04 MY= 0.0000000E+00 MZ= -4.6229579E-04

***TOTAL REACTION LOAD( KN METE ) SUMMARY (LOADING 9 )
SUMMATION FORCE-X = -3.4372086E-13
SUMMATION FORCE-Y = 8.9634104E-14
SUMMATION FORCE-Z = -2.7655702E-14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8.9882103E-12 MY= 7.9606193E-11 MZ= 2.3700223E-06

MAXIMUM DISPLACEMENTS ( CM /RADIANS) (LOADING 9)

| | MAXIMUMS | AT NODE |
|---|---|---|
| X = | 4.03494E-02 | 9 |
| Y = | 4.31414E-02 | 11 |
| Z = | -2.63069E-02 | 5 |
| RX= | -1.18525E-04 | 2 |
| RY= | -1.02281E-05 | 10 |
| RZ= | -1.48566E-04 | 9 |

********** END OF DATA FROM INTERNAL STORAGE **********

226. LOAD LIST 200 TO 214
227. START CONCRETE DESIGN
228. CODE ACI
229. FC 35000 ALL
230. EFACE 0.44 MEMB 1 2 6 7
231. SFACE 0.44 MEMB 1 2 6 7

```
232. EFACE 0.29 MEMB 11 13
233. SFACE 0.29 MEMB 11 13
234. CLB 0.04 MEMB 1 TO 11 13 27 30 TO 36
235. CLS 0.04 MEMB 1 TO 11 13 27 30 TO 36
236. CLT 0.04 MEMB 1 TO 11 13 27 30 TO 36
237. MAXMAIN 16 MEMB 1 TO 11 13 27 30 TO 36
238. MINMAIN 16 MEMB 1 TO 11 13 27 30 TO 36
239. MINMAIN 20 MEMB 4 9 32 35
240. MAXMAIN 20 MEMB 4 9 32 35
241. MINSEC 10 MEMB 1 TO 11 13 27 30 TO 36
242. DESIGN BEAM 1 2 6 7 11 13 25 TO 30 37 38
```

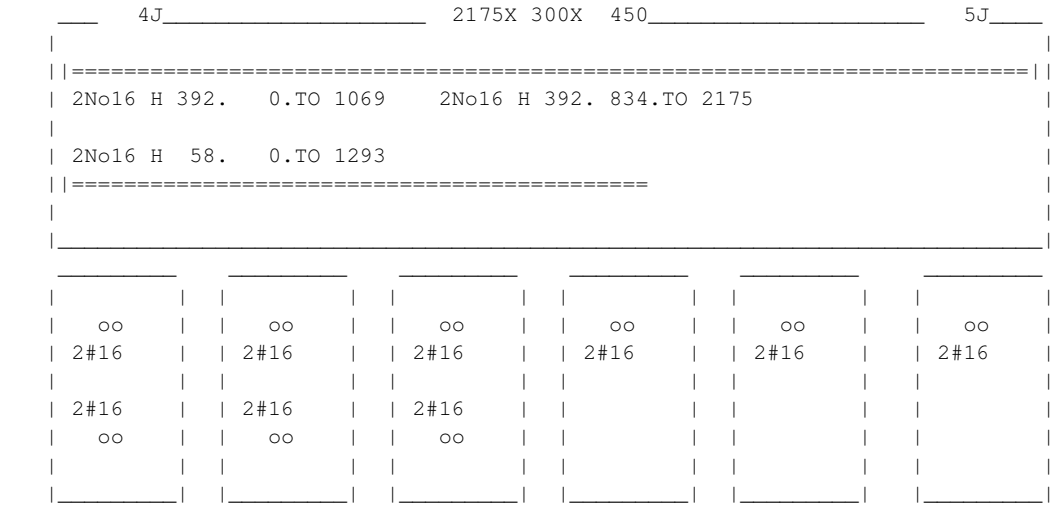
**BEAM NO. 1 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 2175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 58. | 2 - 16MM | 0. | 1293. | YES | NO |
| 2 | 392. | 2 - 16MM | 0. | 1069. | YES | NO |
| 3 | 392. | 2 - 16MM | 834. | 2175. | NO | YES |

**B E A M N O. 1 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 0.92 KNS Vc= 109.93 KNS Vs= 0.00 KNS
Tu= 0.52 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.56 KNS Vc= 202.56 KNS Vs= 0.00 KNS
Tu= 0.20 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



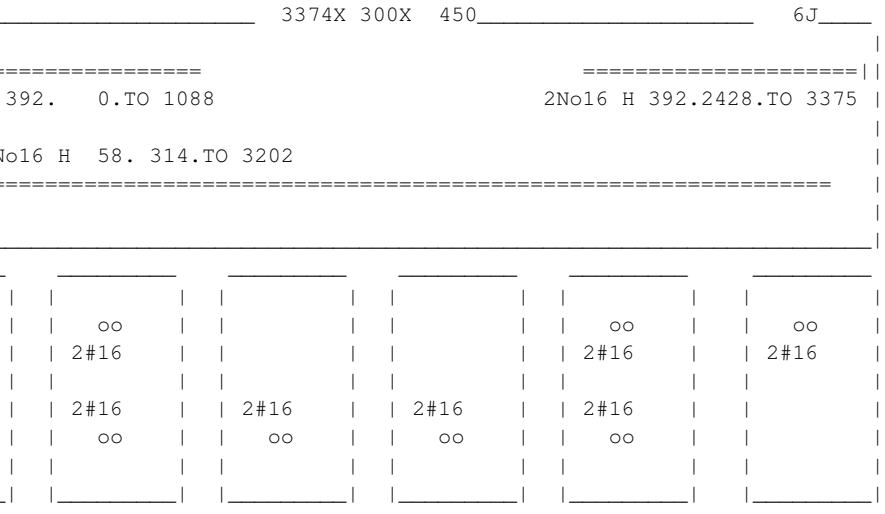
**BEAM NO. 2 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 3375. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 58. | 2 - 16MM | 314. | 3202. | NO | NO |
| 2 | 392. | 2 - 16MM | 0. | 1088. | YES | NO |
| 3 | 392. | 2 - 16MM | 2428. | 3375. | NO | YES |

**B E A M N O. 2 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 13.49 KNS Vc= 113.35 KNS Vs= 0.00 KNS
Tu= 1.14 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 8.09 KNS Vc= 110.56 KNS Vs= 0.00 KNS
Tu= 1.26 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



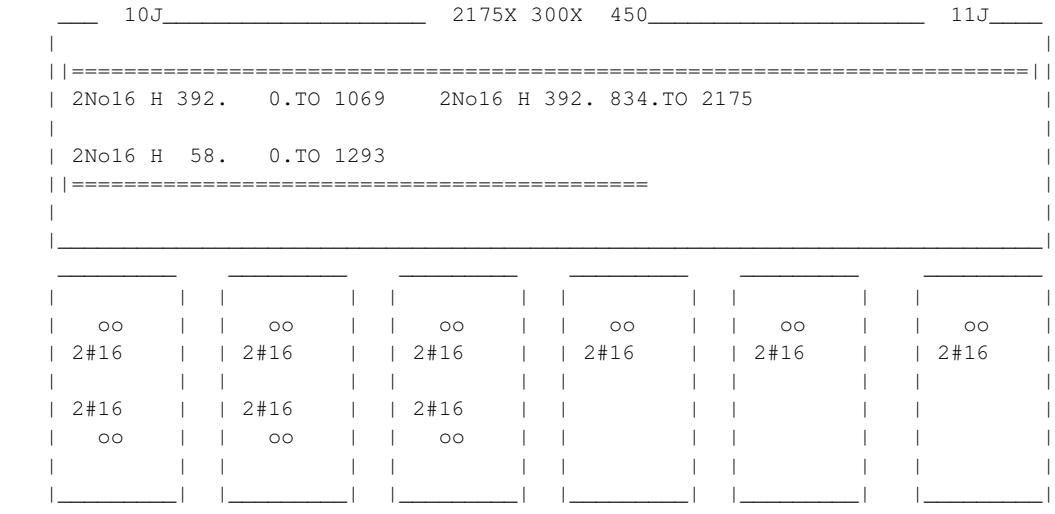
**BEAM NO. 6 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 2175. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 58. | 2 - 16MM | 0. | 1293. | YES | NO |
| 2 | 392. | 2 - 16MM | 0. | 1069. | YES | NO |
| 3 | 392. | 2 - 16MM | 834. | 2175. | NO | YES |

**B E A M N O. 6 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 0.97 KNS Vc= 109.94 KNS Vs= 0.00 KNS
Tu= 0.70 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.51 KNS Vc= 202.55 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



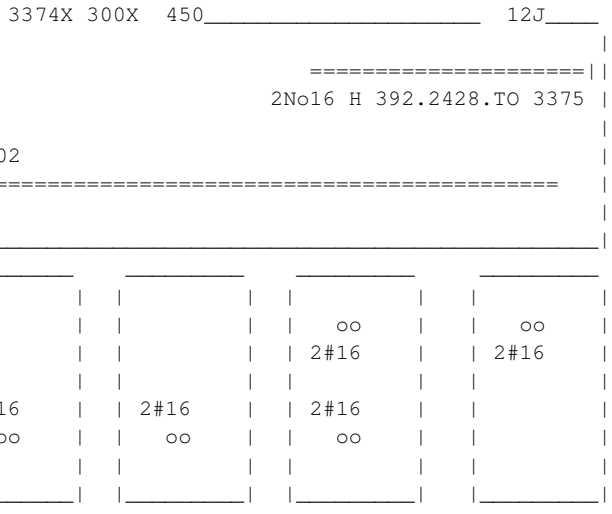
**BEAM NO. 7 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 3375. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 58. | 2 - 16MM | 314. | 3202. | NO | NO |
| 2 | 392. | 2 - 16MM | 0. | 1088. | YES | NO |
| 3 | 392. | 2 - 16MM | 2428. | 3375. | NO | YES |

**B E A M N O. 7 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 13.49 KNS Vc= 113.32 KNS Vs= 0.00 KNS
Tu= 1.02 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 8.09 KNS Vc= 110.56 KNS Vs= 0.00 KNS
Tu= 1.38 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



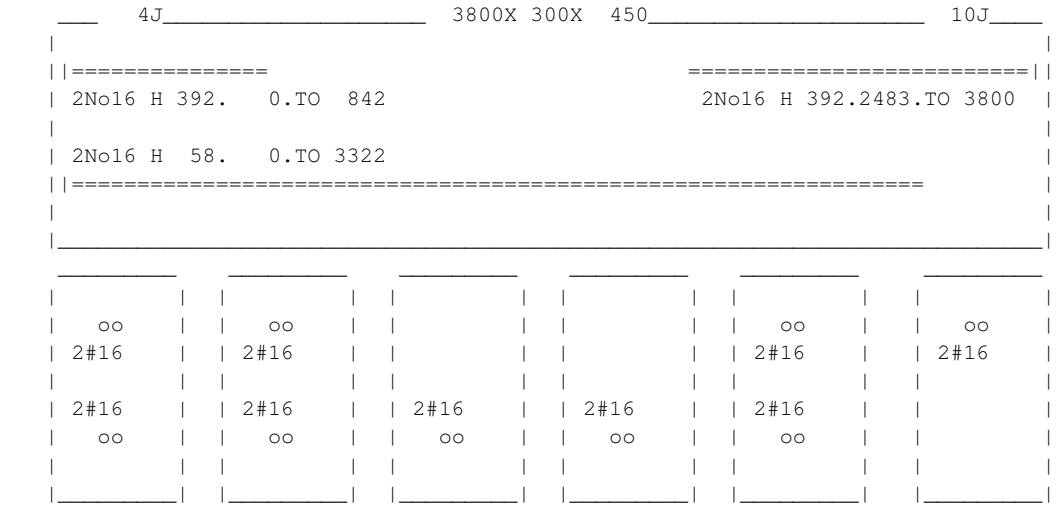
**BEAM NO. 11 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 3800. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 58. | 2 - 16MM | 0. | 3322. | YES | NO |
| 2 | 392. | 2 - 16MM | 0. | 842. | YES | NO |
| 3 | 392. | 2 - 16MM | 2483. | 3800. | NO | YES |

**B E A M N O. 11 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 14.50 KNS Vc= 112.43 KNS Vs= 0.00 KNS
Tu= 1.70 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 16.20 KNS Vc= 114.45 KNS Vs= 0.00 KNS
Tu= 1.71 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



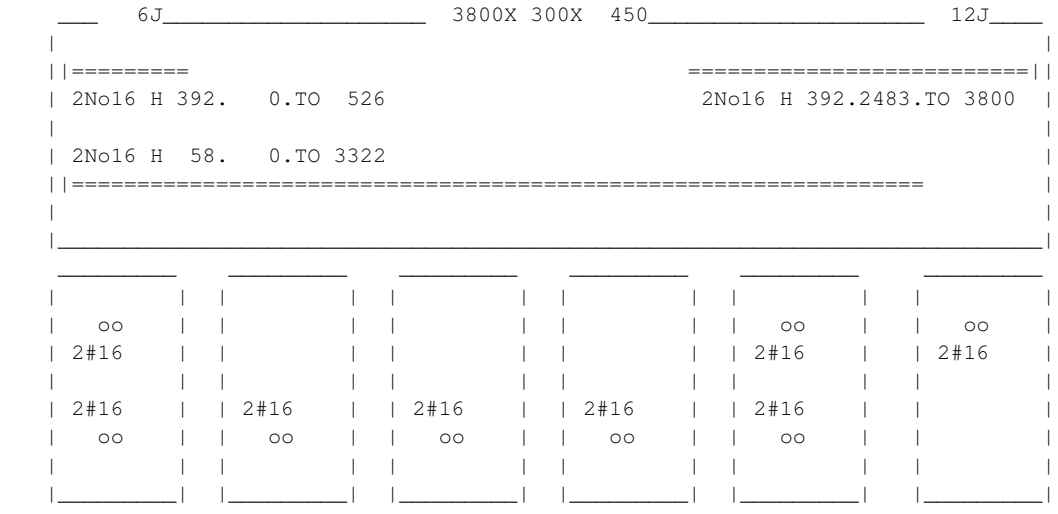
**BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 3800. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 58. | 2 - 16MM | 0. | 3322. | YES | NO |
| 2 | 392. | 2 - 16MM | 0. | 526. | YES | NO |
| 3 | 392. | 2 - 16MM | 2483. | 3800. | NO | YES |

**B E A M N O. 13 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 14.23 KNS Vc= 112.23 KNS Vs= 0.00 KNS
Tu= 1.71 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 16.48 KNS Vc= 115.00 KNS Vs= 0.00 KNS
Tu= 1.70 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



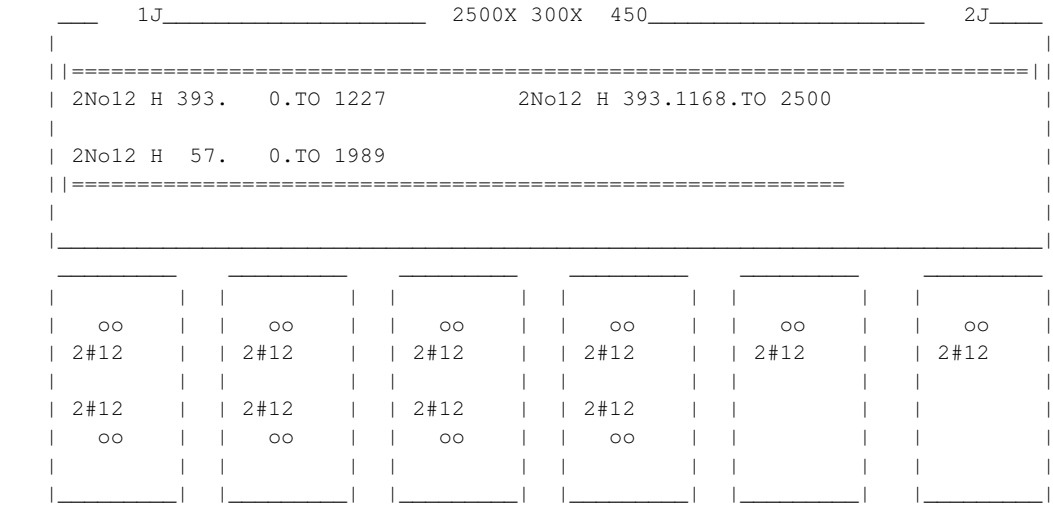
**BEAM NO. 25 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 2500. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 57. | 2 - 12MM | 0. | 1989. | YES | NO |
| 2 | 393. | 2 - 12MM | 0. | 1227. | YES | NO |
| 3 | 393. | 2 - 12MM | 1168. | 2500. | NO | YES |

**B E A M N O. 25 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 13.21 KNS Vc= 203.49 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 18.68 KNS Vc= 112.75 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



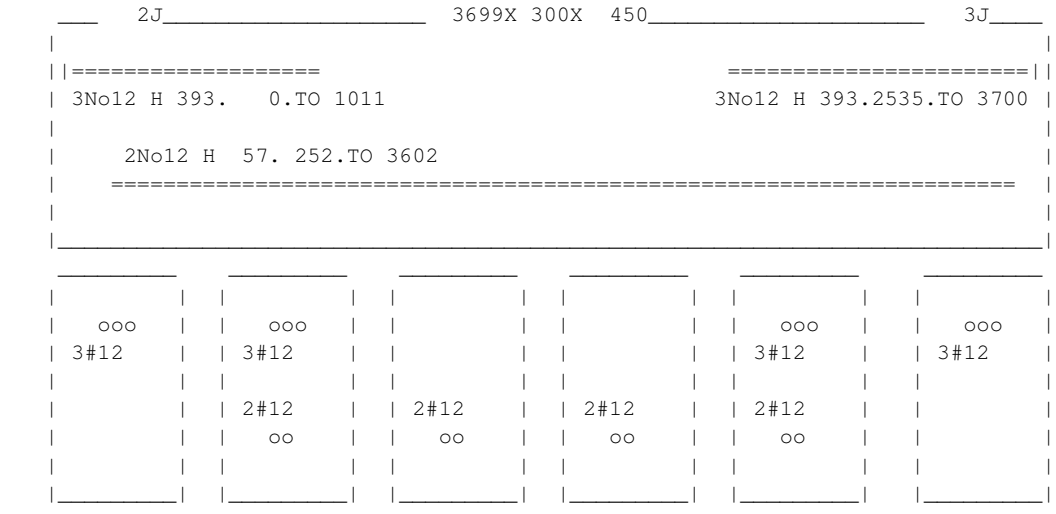
**BEAM NO. 26 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 3700. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 57. | 2 - 12MM | 252. | 3602. | NO | NO |
| 2 | 393. | 3 - 12MM | 0. | 1011. | YES | NO |
| 3 | 393. | 3 - 12MM | 2535. | 3700. | NO | YES |

**B E A M N O. 26 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 28.75 KNS Vc= 113.82 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 25.29 KNS Vc= 117.71 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



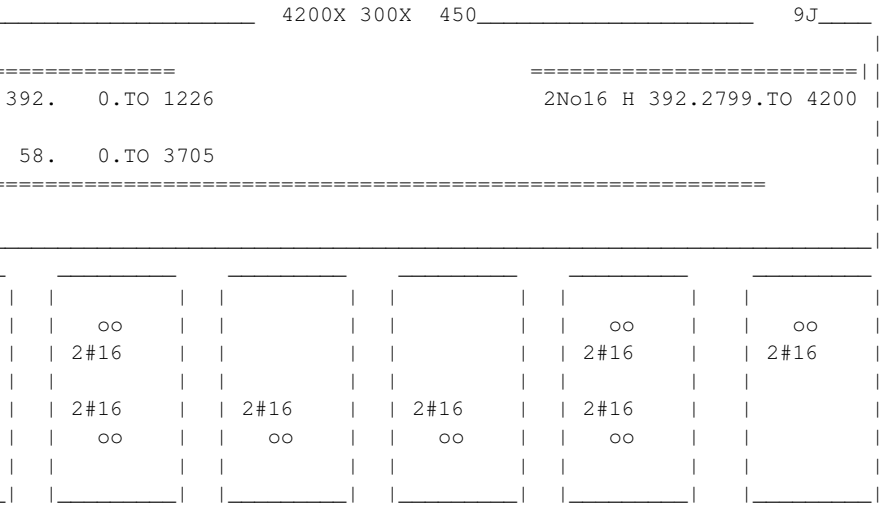
**BEAM NO. 27 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 58. | 2 - 16MM | 0. | 3705. | YES | NO |
| 2 | 392. | 2 - 16MM | 0. | 1226. | YES | NO |
| 3 | 392. | 2 - 16MM | 2799. | 4200. | NO | YES |

**B E A M N O. 27 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 31.47 KNS Vc= 115.15 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 31.67 KNS Vc= 114.97 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



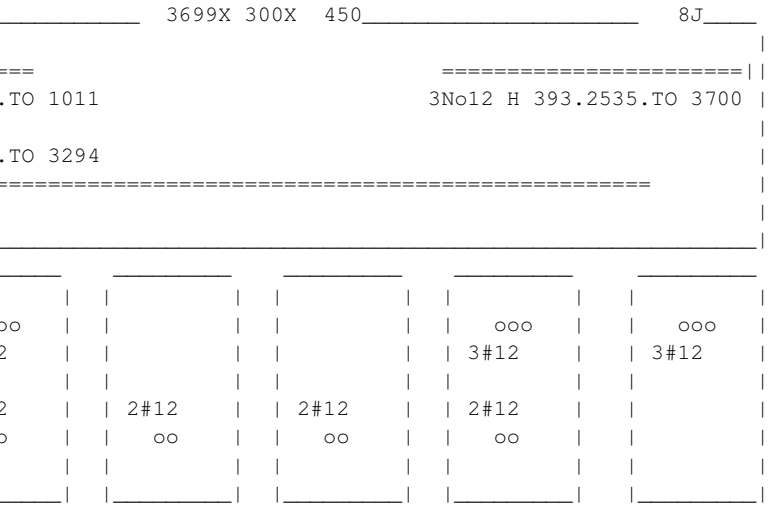
**BEAM NO. 28 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 3700. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 57. | 2 - 12MM | 0. | 3294. | YES | NO |
| 2 | 393. | 3 - 12MM | 0. | 1011. | YES | NO |
| 3 | 393. | 3 - 12MM | 2535. | 3700. | NO | YES |

**B E A M N O. 28 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 25.29 KNS Vc= 117.74 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.75 KNS Vc= 113.82 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



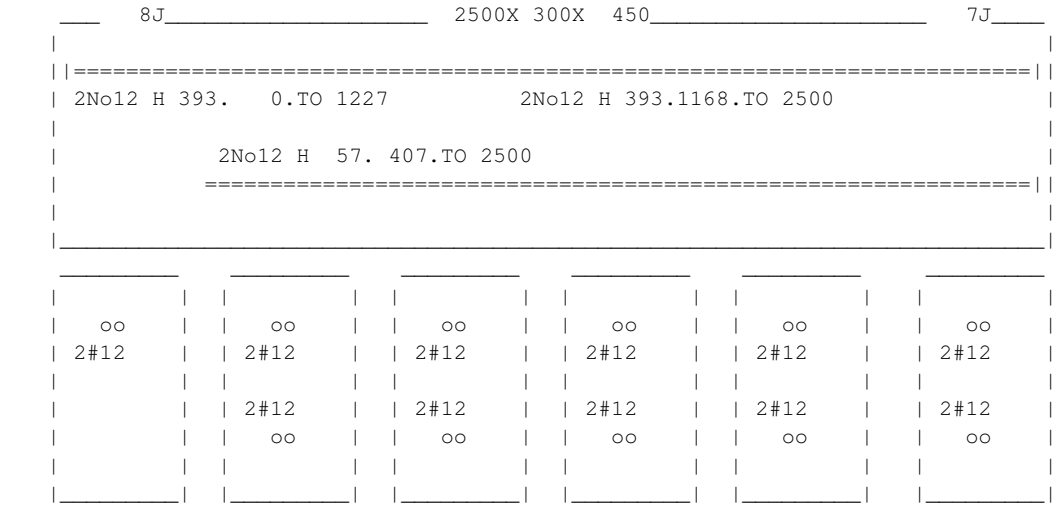
**BEAM NO. 29 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 2500. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 57. | 2 - 12MM | 407. | 2500. | NO | YES |
| 2 | 393. | 2 - 12MM | 0. | 1227. | YES | NO |
| 3 | 393. | 2 - 12MM | 1168. | 2500. | NO | YES |

**B E A M N O. 29 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 18.65 KNS Vc= 112.75 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.24 KNS Vc= 203.49 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



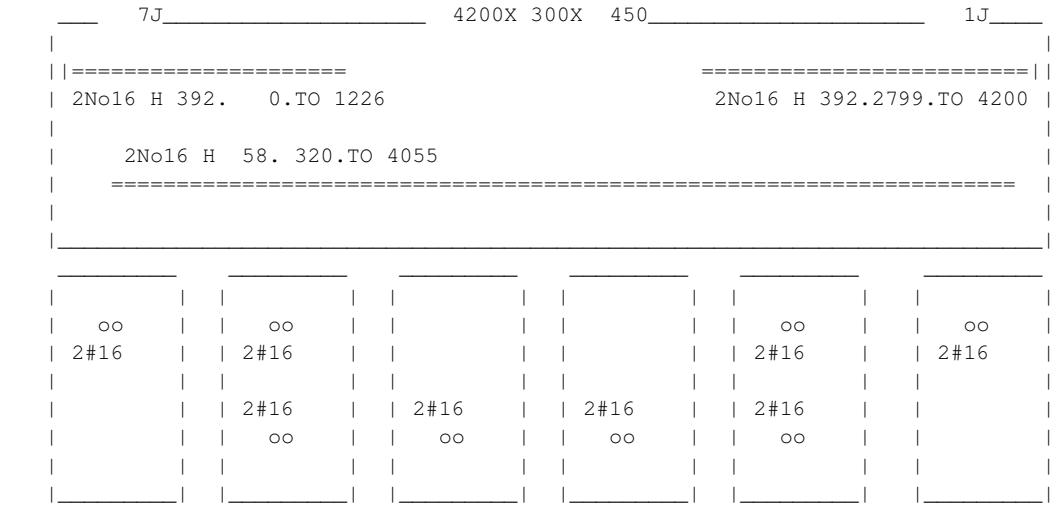
**BEAM NO. 30 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 58. | 2 - 16MM | 320. | 4055. | NO | NO |
| 2 | 392. | 2 - 16MM | 0. | 1226. | YES | NO |
| 3 | 392. | 2 - 16MM | 2799. | 4200. | NO | YES |

**B E A M N O. 30 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 31.68 KNS Vc= 114.96 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 31.46 KNS Vc= 115.17 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



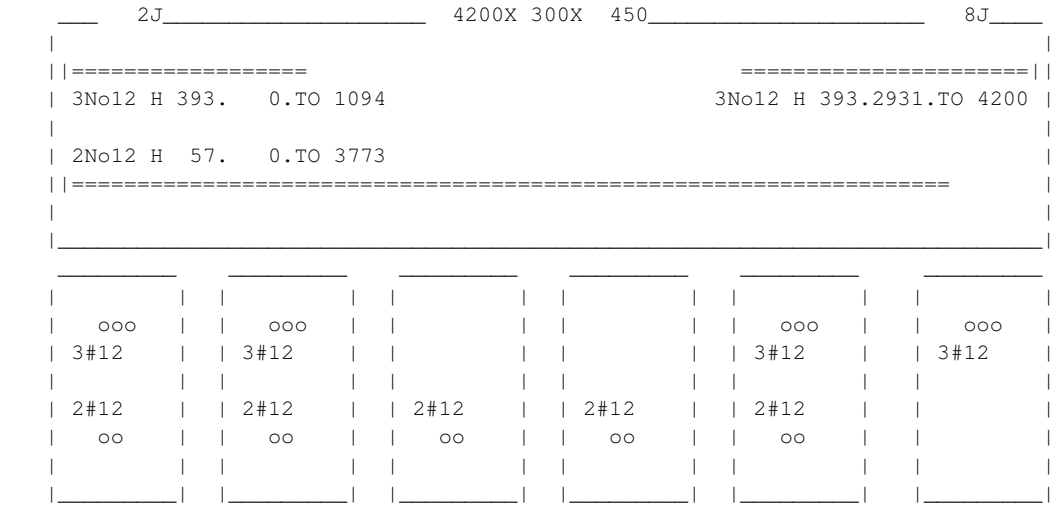
**BEAM NO. 37 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 57. | 2 - 12MM | 0. | 3773. | YES | NO |
| 2 | 393. | 3 - 12MM | 0. | 1094. | YES | NO |
| 3 | 393. | 3 - 12MM | 2931. | 4200. | NO | YES |

**B E A M N O. 37 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 31.60 KNS Vc= 115.63 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 31.67 KNS Vc= 115.57 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.



**==========**

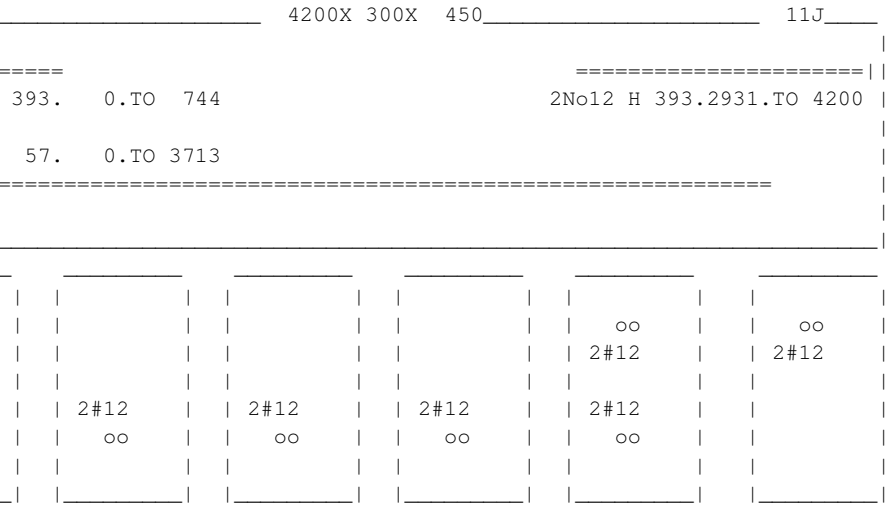
**BEAM NO. 38 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08**

**LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 300. X 450. MMS**

| LEVEL | HEIGHT (MM) | BAR INFO | FROM (MM) | TO (MM) | ANCHOR STA | ANCHOR END |
|---|---|---|---|---|---|---|
| 1 | 57. | 2 - 12MM | 0. | 3713. | YES | NO |
| 2 | 393. | 2 - 12MM | 0. | 744. | YES | NO |
| 3 | 393. | 2 - 12MM | 2931. | 4200. | NO | YES |

**B E A M N O. 38 D E S I G N R E S U L T S - SHEAR**

AT START SUPPORT - Vu= 20.15 KNS Vc= 204.59 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.98 KNS Vc= 116.79 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 200
STIRRUPS ARE NOT REQUIRED.




**********END OF BEAM DESIGN**********

243. DESIGN COLUMN 3 TO 5 8 TO 10 31 TO 36

**========================================================================**

**COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 16 MM | 1.313 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

**========================================================================**

**COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 4 - 20 MM | 1.026 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

**========================================================================**

**COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 16 MM | 1.313 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

**COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 16 MM | 1.313 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

**COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 4 - 20 MM | 1.026 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==========

**COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 16 MM | 1.313 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

**COLUMN NO. 31 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 16 MM | 1.313 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

**COLUMN NO. 32 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 4 - 20 MM | 1.026 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==========

**COLUMN NO. 33 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 16 MM | 1.313 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

**COLUMN NO. 34 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 16 MM | 1.313 | 200 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

==========

**COLUMN NO. 35 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

4 - 20 MM 1.026 200 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

==========

**COLUMN NO. 36 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 350.0 X 350.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1225.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

8 - 16 MM 1.313 200 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

**********END OF COLUMN DESIGN RESULTS**********

244. END CONCRETE DESIGN
245. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 3,2020 TIME= 13:12:43 ****

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*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
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*                http://www.bentley.com                    *
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| | |
|---|---|
| Global Project ID | **900707** |
| Delivery Type | **ENGINEERING PROCUREMENT CONSTRUCTION** |
| Project Name | **MANSOURAH MASSARAH GOLD PROJECT** |
| Customer | **MA'ADEN GOLD AND BASE METALS (MA'ADEN)** |
| Project Country | **KINGDOM OF SAUDI ARABIA** |

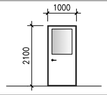
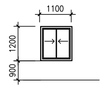
| **Ma'aden Revision No:** | A01 | C01 | 0 | 0 | | |
|---|---|---|---|---|---|---|
| Date | 03-07-2020 | 01-08-2020 | | | | |
| Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | | | | |
| Prepared By | SHD | SHD | | | | |
| Checked | SGC | SGC | | | | |
| Approved | PRI | PRI | | | | |

| Outotec / L&T Revision No.: | 0 | 1 | 2 | 3 | |
|---|---|---|---|---|---|
| Date | 03-07-2020 | 01-08-2020 | | | |
| Revision Text | FIRST ISSUE | ISSUED FOR CONSTRUCTION | | | |
| Prepared By | SHD | SHD | | | |
| Checked | SGC | SGC | | | |
| Approved | PRI | PRI | | | |

**ATTACHMENT 2 Architectutal GA for Power Plant Guard Room**

DRAWING NUMBER: MD - 216 - 49PR-EG - AR-GAD - 3183
REV: A01

## SCHEDULE OF DOOR AND WINDOW OPENINGS

| MKD. | MASONRY OPENING WIDTH | MASONRY OPENING HEIGHT | SILL LEVEL | LINTEL BOTTOM LEVEL | NO. REQD. | DESCRIPTION | REMARKS |
|---|---|---|---|---|---|---|---|
| D1 | 1000 | 2100 | EL+0.000 M | EL+2.100 M | 1 |   | SINGLE PANEL SOLID CORE STEEL DOOR AND FRAME WITH DOOR CLOSER AND VISION PANEL |
| D2 | 750 | 2100 | EL+0.000 M | EL+2.100 M | 1 |   | SINGLE PANEL SOLID CORE STEEL DOOR AND FRAME |
| W1 | 1100 | 1200 | EL+0.900 M | EL+2.100 M | 6 |   | ALUMINIUM GLAZED WINDOW OPENABLE / FIXED |
| W2 | 600 | 600 | EL+ 1.500M | EL+ 2.100M | 1 |   | ALUMINIUM GLAZED FIXED VENTILATOR WITH 6mm THICK TOUGHENED GLASS. |

NOTE:
1. ALL ITEM DETAILS/SPECIFICATION SHALL BE AS PER ITB SPECIFICATION.
2. THE DOOR/WINDOW/ROLLING SHUTTER ELEVATION SHOWN ARE INDICATIVE ONLY. PROCUREMENT & ERECTION AT SITE SHALL BE AS PER ACTUAL MEASUREMENT & APPROVED DETAIL VENDOR DRAWING.

## SCHEDULE OF FLOOR FINISH

| FLOOR | ROOM MKD. | FLOOR FINISH | SKIRTING/DADO | REMARKS |
|---|---|---|---|---|
| EL.+0.000 M. (F.F.L.) | GUARD ROOM | VINYL TILES WITH 150 mm VINYL BASE IN ALL AREAS. | 100 mm HIGH SKIRTING WITH SAME VINYL BASE | |
| | TOILET | 300mmx300mm ANTI SKID CERAMIC TILES | 2100 MM. HIGH GLAZED CERAMIC TILES | |

## SCHEDULE OF WALL FINISH

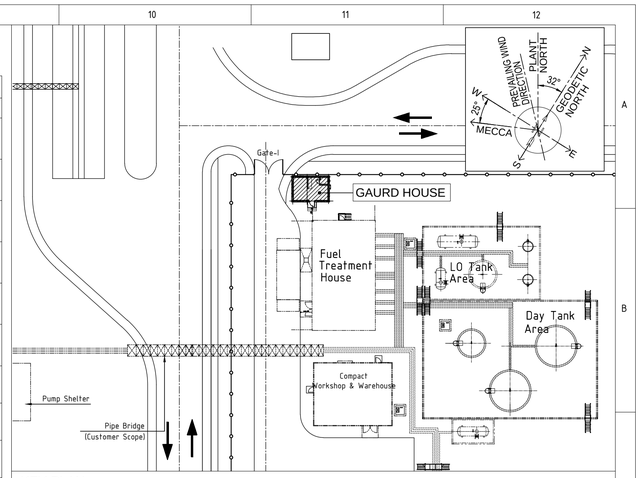
| | AREA | WALL FINISH | PAINT |
|---|---|---|---|
| EXTERIOR WALLS | ALL EXTERIOR WALL | 37.5 MM. PLASTER LAID IN THREE LAYERS. | 2 COATS OF EXTERIOR GRADE EMULSION PAINT OVER 1 COAT OF VINYL EMULSION PRIMER WITH 0.065 mm DRY FLIM THICKNESS |
| INTERIOR WALLS | GUARD ROOM | 25 MM. PLASTER LAID IN TWO LAYERS. (ABOVE SKIRTING) | 2 COATS OF INTERIOR QUALITY ACRYLIC EMULSION PAINT OVER 1 COAT OF LATEX PRIMER |
| | TOILET | | |
| CEILING SURFACE | GUARD ROOM | NO PLASTER | ACOUSTICAL FALSE CEILING |
| | TOILET | | MOISTURE RESISTANT FALSE CEILING |

## SCHEDULE OF HVAC OPENINGS

| MKD. | MASONRY OPENING | ELEVATION | NO. REQD. | DESCRIPTION | REMARKS |
|---|---|---|---|---|---|
| V1 | 600 DIA | CL.O.OP. AT EL+2.55M | 1 | WALL OPENING FOR EXHAUST FAN | WALL OPENINGS ARE CLEAR & TENTATIVE & SHALL BE FINALIZED DURING DETAIL ENGINEERING, FOR DETAIL OF ALL OPENINGS & CURB ANGLE REFER RELEVANT HVAC DRAWINGS. |

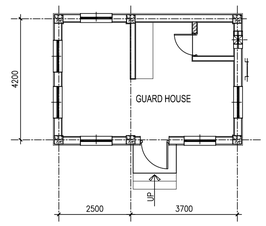
## LEGEND AND DESCRIPTION

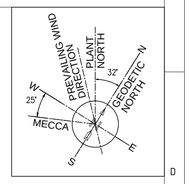
| WALL | FINISHES | SAMPLE |
|---|---|---|
| W01 | 250 MM THK. CMU BLOCK WALL WITH INSULATION AND 38 MM THK. PLASTER EXTERNAL AND 25 MM THK.PLASTER INSIDE. | |
| W02 | 150 MM THK. CMU BLOCK WALL WITH 25 MM, THK. PLASTER FINISHING FAIR FACED ON THE DRY SIDE AND GLAZED CERAMIC TILING ON TOILET SIDE. | |
| RCC | RCC COLUMN / BEAM / SLAB | |
| B/W | BLOCKWORK IN ELEVATION | |
| PCC | LEAN CONCRETE | |
| INS | INSULATION | |
| | COMPACTED FILL | |
| | HOLLOW CORE SLAB | |
| | GYPSUM BOARD FALSE CEILING (START POINT) | |
| | GYPSUM BOARD MOISTURE PROOFING FALSE CEILING | |

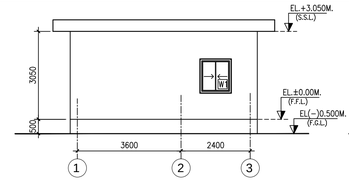



KEY PLAN
SCALE 1:500

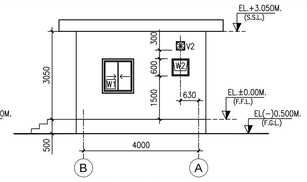
BUILDING KEY PLAN



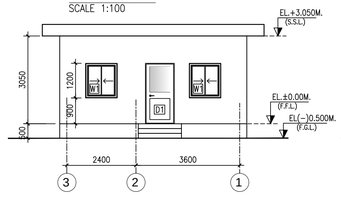



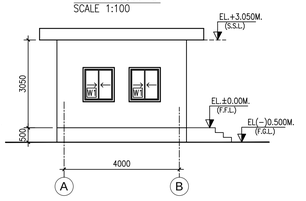

ELEVATION - C
SCALE 1:100



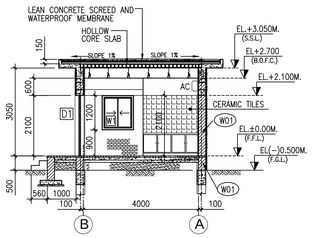

ELEVATION - D
SCALE 1:100



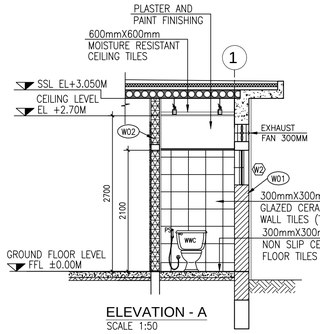

ELEVATION - A
SCALE 1:100



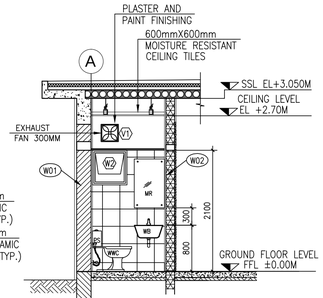

ELEVATION - B
SCALE 1:100



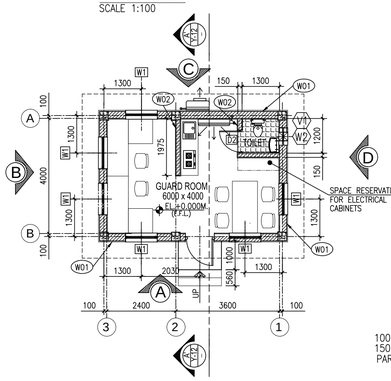

SECTION A - A
SCALE 1:100



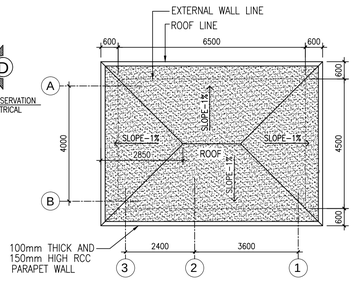

ELEVATION - A
SCALE 1:50



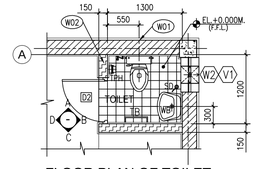

ELEVATION - B
SCALE 1:50



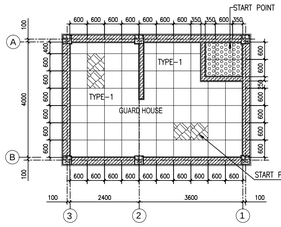

PLAN AT EL.(+)0.000M (F.F.L.)
SCALE 1:100



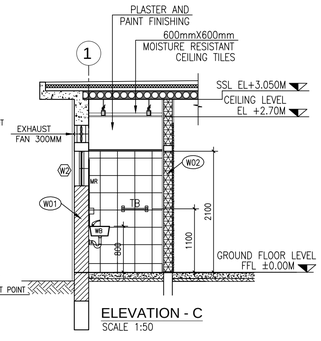

PLAN AT ROOF LEVEL
SCALE 1:100



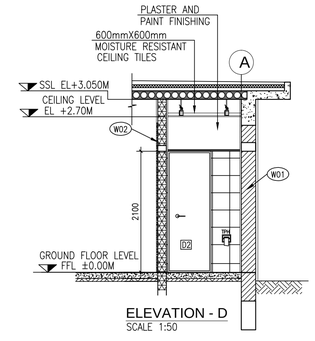

FLOOR PLAN OF TOILET
SCALE 1:50




REFLECTED CEILING PLAN AT EL(+)2.700M (B.O.CELING)
SCALE 1:75




ELEVATION - C
SCALE 1:50




ELEVATION - D
SCALE 1:50

### ABBREVATIONS / LEGENDS

| | |
|---|---|
| FGL | FINISH GRADE LEVEL |
| FFL | FINISH FLOOR LEVEL |
| TYP. | TYPICAL |
| CMU | CONCRETE MASONRY UNIT |
| SSL | STRUCTURAL SLAB LEVEL |
| TOP | TOP OF PARAPET |
| DN | DOWN |
| B.O.F.C. | BOTTOM OF FALSE CEILING |
| RCC | REINFORCEMENT CEMENT CONCRETE |
| PCC | PLAIN CEMENT CONCRETE |

### TOILET SANITARY LEGEND

| SYMBOL | ABBREV. | DESCRIPTION |
|---|---|---|
| | WWC | WESTERN WATER CLOSET |
| | PS | PERENNIAL SPRAY |
| | WB | LAVATORY, FREESTANDING WITH SMART MIXER |
| | TPH | TOILET PAPER HOLDER |
| | MR | FACIAL MIRROR |
| | RH | ROBE HOOK |
| | SP | START POINT OF FLOOR TILES AND CEILING |
| | PICH | PIPE CHASE |
| | ELV T | ELEVATION TARGET (ARCHITECTURAL) |
| | SD | SOAP DISH |
| | TB | TOWEL BAR |

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER & LEVELS ARE IN METER UNLESS OTHERWISE STATED.
2. ALL DWG. IN 1:100 SCALE UNLESS OTHERWISE STATED.
3. ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH ALL OTHER ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
4. FOR GRADED FINISHED FLOOR LEVEL (F.F.L) REFER DRAWING NUMBER MD-216-1000-EG-PPD-1002, PLOT PLAN GENERAL PROCESS PLANT LAYOUT.
5. FOR ALL HARDSCAPE AND LANDSCAPE ELEMENTS AND DETAILS REFER TO SEPARATE LANDSCAPE DRAWINGS.
6. FOR BUILDING SITE SETTING OUT PLAN AND DATA, SEE SURVEYORS/CIVIL DRAWINGS.
7. FOR BUILDING ELEVATION, SEE GRADING PLAN.
8. FOR DETAILS OF ALL ELECTRICAL OPENINGS REFER RELEVANT ELECTRICAL DRAWINGS. THE LIGHT FIXTURE SHOWN ARE TENTATIVE, SHALL BE FINALISED AS PER ELECTRICAL DRAWING.
9. FOR DETAILS OF ALL HVAC OPENINGS & CURB ANGLE REFER RELEVANT HVAC DRAWINGS.
10. ALL OPENING SIZES ARE CLEAR MASONRY OPENING.

| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|
| MD-216-1000-EG-ST-CRT-1001 | CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA | | | | | | | | |
| MD-216-XXXX-XX-XX-XXX-XXXX | GAURD HOUSE - PLAN, SECTION AND ELEVATION | A01 | 29/06/2020 | ISSUED FOR APPROVAL | MD | DRO | DRO | SJM | |

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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| DRAWN | MD | 15/06/2020 | |
| CHECKED | DRO | 18/06/2020 | |
| ENGINEER | DRO | 26/06/2020 | |
| APPROVED | SJM | 29/06/2020 | |
| EMPLOYER | | | |
| SCALE | 1:100 | A1 | |

LARSEN & TOUBRO DRAWING NUMBER: O19045_008_A_G_25048_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 __GAU01 __WG01__MTL01 __49004 — DOCUMENT ID: OU100665768 — REVISION: 0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
POWER GENERATION : GUARD ROOM
PLAN, ELEVATION, SECTIONS & DETAILS

DRAWING NUMBER: MD - 216 - 49PR- EG- AR- GAD - 3183 — REV: A01